

Impossibility Theorems for Feature Attribution

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Abstract

Despite a sea of interpretability methods that can produce plausible explanations, the field has also empirically seen many failure cases of such methods. In light of these results, it remains unclear for practitioners how to use these methods and choose between them in a principled way. In this paper, we show that for moderately rich model classes (easily satisfied by neural networks), any feature attribution method that is complete and linear—for example, Integrated Gradients and SHAP—can provably fail to improve on random guessing for inferring model behaviour. Our results apply to common end-tasks such as characterizing local model behaviour, identifying spurious features, and algorithmic recourse. One takeaway from our work is the importance of concretely defining end-tasks: once such an end-task is defined, a simple and direct approach of repeated model evaluations can outperform many other complex feature attribution methods.

1 Introduction

Feature attribution methods are commonly used to answer local counterfactual questions about machine learning models; that is, questions about a model’s behaviour $f(x)$ near a particular example x . For example, the goal of *algorithmic recourse* is to determine what changes to x a user should make to change the model’s prediction, such as whether raising an applicant’s credit score would affect the model’s predicted probability of loan default. Similarly, the goal of *spurious feature identification* is to determine whether features that should be ignored by the model actually affect $f(x)$; for example, does the model use a watermark on an X-ray to predict the probability of disease? These feature attribution methods generally fall into one of two categories: first, local approximations of $f(x)$ in a sufficiently small neighbourhood around x , such as taking the gradient of f near x [36] or using more sophisticated approximations like SmoothGrad [37] and LIME [33]; and second, methods that incorporate how the model behaves on a *baseline distribution* μ over examples that might be far from the example of interest, as in methods like SHAP [23] and Integrated Gradients [41].

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The failure modes of the first category—local methods like taking the gradient of f —are well understood. For example, if one is interested in recourse directions that extend outside of the neighbourhood used by a local method, then the resulting feature attribution may not reliably answer such questions. These failures have spurred the development of methods in the second category, which are often motivated as provably satisfying properties such as completeness and linearity. These methods—including SHAP and Integrated Gradients (IG)—are commonly considered more reliable and are widely used to answer local counterfactual questions across many important applications, including clinical trials [22], medical alerts in the ICU [48], cancer diagnosis [49] and prognosis [34], and chemical binding [24]. As a concrete example, consider the clinical trial setting of Liu et al. [22], where the counterfactual model behaviour of interest is whether removing certain eligibility criteria for participation in a clinical trial will change the efficacy of the trial (measured by the hazard ratio for patient survival). The authors infer this counterfactual model behaviour by concluding that “Shapley values close to zero [...] correspond to eligibility criteria that had no effect on the hazard ratio of the overall survival.”

In this work, we show that these common conceptions can be misleading: complete and linear methods are in fact often less reliable than simpler methods—such as taking the gradient—at answering local counterfactual questions. Our main result is that for *any* feature attribution method that satisfies the completeness and linearity axioms, users cannot generally do better than random guessing for end-tasks such as algorithmic recourse and spurious feature identification. Specifically, there are uncountably many pairs of models that share a feature attribution yet have arbitrarily different counterfactual model behaviour. Conversely, for every pair of distinct feature attribution values, there are uncountably many pairs of models that match these feature attributions and yet have identical counterfactual model behaviour. Furthermore, unlike simple local methods such as taking the gradient, complete and linear methods remain unreliable even if we restrict our attention to infinitesimally small neighborhoods around the example of interest x , because they remain sensitive to how the model behaves on the baseline μ that might be far from x . Intuitively, there are two issues at play: (a) the completeness axiom requires attributions to sum to something meaningful, which is generally not well-aligned with answering questions about counterfactual model behaviour, and (b) the reliance on a baseline introduces additional failure modes due to how the model behaves far from the example of interest.

Our results have direct implications for using complete and linear methods to answer questions about counterfactual model behaviour. For instance, positive feature attribution does **not**, in general, imply that increasing the feature will increase the model output. Similarly, zero feature attribution does **not**, in general, imply that the model output is insensitive to changes in the feature (see Figure 1 for a visualization of false implications). Beyond our general impossibility results, these methods can also fail to infer counterfactual model behaviour *on average* over models. We show this both theoretically (over a distribution of polynomial models in Section 3.5) and empirically (on eight standard datasets, comprising tabular features and image classification, in Section 4).

In light of these results, we end by discussing how, in the absence of reliable methods—that is, outside of special settings, such as when the model is linear or when the counterfactual is

How is counterfactual model behaviour related to end-tasks that users may care about in real-world applications? We consider two common applications in the literature, algorithmic recourse and spurious feature identification, to motivate inferring counterfactual model behaviour. For algorithmic recourse, we consider the task of determining whether increasing or decreasing a given feature would be more beneficial. For spurious feature identification, we consider the task of distinguishing if the model output is sensitive to local perturbations in the feature. We formalize both of these tasks in Section 3.4.2.

2.2 Framing the Problem: Hypothesis Testing

We formulate learning counterfactual model behaviour as a hypothesis testing problem, which gives us a framework to measure and compare the performance of different feature attribution methods. Following the usual nomenclature, the user’s goal is to determine whether the model has certain behaviour: the *null hypothesis*. This behaviour is contrasted with some different, plausible model behaviour: the *alternate hypothesis*. The null hypothesis and alternate hypothesis should encode necessary (but potentially not sufficient) questions that must be answered to succeed at the task of inferring counterfactual model behaviour. For example, for recourse the user must be able to infer if the model is increasing or decreasing, while for spurious features the user must be able to infer if the model is sensitive or insensitive to perturbations of a feature. After collecting information about the model using a feature attribution method, the user will then conduct a hypothesis test and either *reject* or *fail to reject* the null hypothesis.

Formally, the null and alternate hypotheses define subsets $\mathcal{F}^{(0)} \subseteq \mathcal{F}$ and $\mathcal{F}^{(1)} \subseteq \mathcal{F}$, with

$$\begin{aligned} H^{(0)} &: f \in \mathcal{F}^{(0)} \\ H^{(1)} &: f \in \mathcal{F}^{(1)}. \end{aligned}$$

Here, $\mathcal{F}^{(0)}$ contains $f^{(0)}$ ’s that locally agree with the possible $g^{(0)}$ ’s encoding counterfactual model behaviour, such as all *increasing* models. Similarly, $\mathcal{F}^{(1)}$ contains all models with desired contrasting model behaviour. While it must be the case that these sets have no overlap ($\mathcal{F}^{(0)} \cap \mathcal{F}^{(1)} = \emptyset$), it is often the case that they do not exhaustively contain all models ($\mathcal{F}^{(0)} \cup \mathcal{F}^{(1)} \neq \mathcal{F}$). A *feature-attribution hypothesis test* is any way for the user to draw their conclusion for the hypothesis test solely on the output of a feature attribution method at an example. Formally, this is any function

$$\mathbf{h} : \mathbb{R}^{p \times q} \rightarrow [0, 1].$$

The user may rely on external randomness to decide whether to accept or reject the null hypothesis: the output of $\mathbf{h}$ is *the probability that the user rejects the null hypothesis*.

2.3 Evaluating Hypothesis Tests

Most generally, a *feature attribution method* is any function

$$\Phi : \mathcal{F} \times \mathcal{P}(\mathcal{X}) \times \mathcal{X} \rightarrow \mathbb{R}^{p \times q},$$

where for any space $\mathcal{Z}$, $\mathcal{P}(\mathcal{Z})$ denotes the set of all probability measures on $\mathcal{Z}$. This captures the most common properties of existing definitions in the literature, facilitating their comparison. First, every feature attribution method must take as input the model $f \in \mathcal{F}$ that

the user wishes to understand. Second, the user can specify an example $x \in \mathcal{X}$ to “localize” the feature attribution. In contrast to “global” methods, which seek to explain the effect of features on the model output across a population, localization allows the user to seek counterfactual model behaviour relative to a specific example. Finally, many methods rely heavily on the presence of a *baseline* $\mu \in \mathcal{P}(\mathcal{X})$ to incorporate model behaviour from other examples into computing feature attribution. This baseline may encode the training data distribution (e.g., this is what SHAP does), concentrate entirely on a single example (e.g., this is what IG does), or take some other form specific to the task of interest.

We concretely define a few methods widely used in practice and the focus of our analysis. Gradient [36] can be written as $\Phi(f, \mu, x) = \nabla f(x)$ (notice this is constant with respect to choice of μ). Similarly, IG [41] can be written as

$$\Phi(f, \mu, x)_j = \mathbb{E}_{X \sim \mu} \left[(x_j - X_j) \int_0^1 \nabla_j f(X + \alpha(x - X)) d\alpha \right],$$

where μ is often taken to be a pointmass. For a more complete discussion of precise definitions and properties (including the definition of SHAP), see Appendix A.

The goal of our work is to see if certain feature attribution methods can reliably be used to conduct the hypothesis tests described above. What are the right metrics to measure the success or failure? Classically [47], the quality of a hypothesis test is determined by its *specificity*, which is the probability that the user fails to reject the null hypothesis whenever the model satisfies the null hypothesis (a *true negative*, or $1 -$ probability of a type-II error), and its *sensitivity*¹, which is the probability that the user rejects the null hypothesis whenever the model satisfies the alternate hypothesis (a *true positive*, or $1 -$ probability of a type-I error). In particular, for a fixed feature attribution method Φ , baseline $\mu \in \mathcal{P}(\mathcal{X})$, example $x \in \mathcal{X}$, and null and alternate hypotheses $\mathcal{F}^{(0)}, \mathcal{F}^{(1)} \subseteq \mathcal{F}$,

$$\begin{aligned} \text{Spec}_{\Phi, \mu, x}(\mathbf{h}) &= \inf_{f \in \mathcal{F}^{(0)}} [1 - \mathbf{h}(\Phi(f, \mu, x))] \quad \text{and} \\ \text{Sens}_{\Phi, \mu, x}(\mathbf{h}) &= \inf_{f \in \mathcal{F}^{(1)}} \mathbf{h}(\Phi(f, \mu, x)). \end{aligned}$$

For every hypothesis test, it is trivial to construct *some* model with accurate inference; our measure of performance instead asks if the hypothesis test does well for *all* models of interest (mathematically, this is the role of the $\inf$ ²). The goal of the user is to simultaneously maximize specificity and sensitivity (both take values in $[0, 1]$).

How does this framework differ from other hypothesis testing, such as t-testing? Our definitions of specificity and sensitivity are the exact analogues of the definitions used in classical statistical hypothesis testing. There, $\mathcal{F}^{(0)}$ denotes the true data-generating model rather than the learned model, but the requirement that the test does well *for all* such models is still enforced. As a concrete example, if the true data-generating model is assumed to be

¹The interpretability literature sometimes uses the term *sensitivity* to refer to the effect of perturbations on feature attributions [46]. We use only the hypothesis testing definition above.

²The reader may replace “sup” with “max” and “inf” with “min” without change of meaning; the difference is only that the spaces may not be compact and hence the arg max and arg min may not be achieved.

$y = f_\beta(x) = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ where ε is standard Gaussian noise, then a canonical hypothesis test uses the null hypothesis $\mathcal{F}^{(0)} = \{f_\beta : \beta_1 = 0\}$ to test if the first coefficient is non-zero. The standard requirement of using a hypothesis test $\mathbf{h}$ based on a test statistic with p-value less than α is exactly the requirement that

$$\inf_{f \in \mathcal{F}^{(0)}} \mathbb{E}_{y_{1:n} \sim f} [1 - \mathbf{h}(y_{1:n})] \geq 1 - \alpha.$$

Since we have no randomness over the data (the model is already learned, not retrained), our definition is a deterministic version of this standard worst-case hypothesis test.

2.4 Notation

We end this section with the standard notation used throughout the paper.

For any integer i , let $[i] = \{1, \dots, i\}$ and e_i be the i th standard basis vector. For any set $\mathcal{A}$, let $\mathcal{A}^c$ denote its complement; also, let $\emptyset$ denote the empty set. For any example $x \in \mathcal{X}$, let x_j denote the j th element/feature.

Often, we want to evaluate $f(z)$ where $z \in \mathbb{R}^p$ is defined such that

$$z_\ell = \begin{cases} x_\ell & \ell \neq j \\ x'_\ell & \ell = j \end{cases}$$

for some examples $x, x' \in \mathcal{X}$ and feature $j \in [p]$. That is, replacing the j th feature of x with x'_j . For clarity, we overload the notation of f and write $f(x_{[p] \setminus \{j\}}, x'_j) = f(z)$.

We use shorthand vector notation, letting $x^{(1:n)} = (x^{(1)}, \dots, x^{(n)}) \in \mathcal{X}^n$. Similarly, we write $f(x^{(1:n)}) = (f(x^{(1)}), \dots, f(x^{(n)}))$. For any matrix M , let M_j denote the j th row of M . We use capital letters to denote random variables, and for any $\mu \in \mathcal{P}(\mathcal{X})$ and $f : \mathcal{X} \rightarrow \mathcal{Y}$, let

$$\mathbb{E}_{X \sim \mu} [f(X)] = \int f(x) \mu(dx),$$

with similar notation for variances, covariances, etc. For any $j \in [p]$, let μ_j denote the j th marginal distribution of μ . Note that, unless specifically mentioned, we make no assumption that the features are independent under μ .

3 Impossibility Theorems

Our main result is that **for mildly rich model classes, it is impossible to conclude that the user does better than random guessing at inferring counterfactual model behaviour using common feature attribution methods without strong additional assumptions on the learning algorithm or data distribution.** We also apply this result to two common end-tasks, showing that under these conditions, it is impossible to conclude that the user does better than random guessing for algorithmic recourse and spurious feature identification using these feature attribution methods. Finally, we show that even for very simple models (so our richness condition does not hold), these feature attribution methods provably still incorrectly infer counterfactual model behaviour with significant probability.

To characterize the common feature attribution methods we consider, we begin by defining two commonly used properties of such methods: *completeness* and *linearity*. Informally, completeness requires that the individual feature attributions sum to the difference of the model output from the baseline, while linearity requires that if the model has no feature interactions then the attributions of a given feature should be equivalent to considering that feature on its own.

Definition 3.1 (Complete). *A feature attribution method Φ is complete if and only if for all models $f \in \mathcal{F}$, baselines $\mu \in \mathcal{P}(\mathcal{X})$, and examples $x \in \mathcal{X}$,*

$$\sum_{j \in [p]} \Phi(f, \mu, x)_j = f(x) - \mathbb{E}_{X \sim \mu} f(X).$$

Definition 3.2 (Linear). *A feature attribution method Φ is linear if and only if for all collections of functions $f^{(1)}, \dots, f^{(p)} : \mathbb{R} \rightarrow \mathcal{Y}$, if $f(x) = \sum_{j \in [p]} f^{(j)}(x_j) \in \mathcal{F}$ then for all baselines $\mu \in \mathcal{P}(\mathcal{X})$, examples $x \in \mathcal{X}$, and features $j \in [p]$*

$$\Phi(f, \mu, x)_j = \Phi(f_j, \mu_j, x_j).$$

Two of the most common feature attribution methods, SHAP [23] and Integrated Gradients [41], are complete and linear (for precise definitions and statements, see Appendix A).

3.1 Assumptions on the Model

Our theoretical results rely on two mild assumptions about the features and the model class. These assumptions are typically satisfied in the cases we are interested in: neural networks applied beyond toy problems. This implies that to do better than random guessing with common feature attribution methods, the user must necessarily introduce more structure than this standard setting.

Assumption 1 (Informal). This assumption has two parts. First, we require that $\mathcal{X}$ is not degenerate near x ; i.e., there exists an interval around x on which we can study counterfactual model behaviour. If this were not true, it would not make sense to ask about the model behaviour “near x ” (encoded by δ as described in Section 2.1). In particular, we restrict ourselves to continuous input spaces. For sufficiently rich discrete input spaces (i.e., taking more than a few distinct values), our results could be proved in the same way with more notational overhead.

Second, we require that the baseline has support outside of this interval. This assumption is mild in practice. For example, when the baseline is the training distribution, there only needs to be a single training example that does not fall in the interval above. When the baseline concentrates on a single example, this example must fall outside of the interval; in this case, the baseline example usually corresponds to setting all features to zero, or all pixels to black, and hence is sufficiently far from any x of interest. We formalize this as follows.

Assumption 1. *For any example $x \in \mathcal{X}$, feature $j \in [p]$, radius $\delta > 0$, and baseline $\mu \in \mathcal{P}(\mathcal{X})$, we say that the present assumption holds if there exist $x_j^L, x_j^R \in \mathbb{R}$ such that*

$$[x_j - \delta, x_j + \delta] \subseteq (x_j^L, x_j^R) \subseteq \{x'_j : x' \in \mathcal{X}\}$$

and

$$\mu_j((x_j^L, x_j^R) \setminus [x_j - \delta_j, x_j + \delta_j]) > 0.$$

Assumption 2 (Informal). General impossibility results are unobtainable; for example, most feature attribution methods correctly identify the model for linear models. In other words, to derive an implication similar to ours (that the feature attribution does not provide information about counterfactual model behaviour) one must impose some constraint on the model class. The purpose of this assumption is to define how rich a model would have to be for our result to hold. Note that this assumption includes standard machine learning architectures such as neural networks with reasonable complexity.

In particular, we require that the model class $\mathcal{F}$ can represent sufficiently many piecewise linear extensions of the local counterfactual model behaviour (encoded by g as described in Section 2.1) that the user wishes to infer. This is naturally satisfied by ReLU networks: every ReLU network is a piecewise linear function, and any ReLU network of logarithmic (in dimension) depth or polynomial (in number of pieces) size can exactly replicate any piecewise linear function [for precise bounds, see 5, 42, 7]. This does not preclude the model class from being arbitrarily richer than piecewise linear functions.

To precisely describe Assumption 2, we need some standard notation. For any neighbourhood $\mathcal{B} \subseteq \mathcal{X}$ and model $f : \mathcal{X} \rightarrow \mathcal{Y}$, let $f|_{\mathcal{B}}$ be the model restricted to have the domain $\mathcal{B}$; that is, $f|_{\mathcal{B}} = (\mathcal{B} \ni x \mapsto f(x))$. Similarly, let $\mathcal{F}|_{\mathcal{B}} = \{f|_{\mathcal{B}} : f \in \mathcal{F}\}$ denote the collection of all such restrictions. We say that f is a *m-piecewise linear function on $\mathcal{B} \subseteq \mathcal{X}$* if there exist closed, convex polytopes $V_1, \dots, V_m$ that partition $\mathcal{B}$ such that $f|_{V_j}$ is linear for each $j \in [m]$ and f is continuous. This is a standard definition, for example, see Definition 1 of Chen et al. [7]. Let $\mathcal{L}^m|_{\mathcal{B}}$ denote the set of all m -piecewise linear elements of $\mathcal{B} \rightarrow \mathcal{Y}$. We place the following assumption on the minimum richness of the model class.

Assumption 2. For any example $x \in \mathcal{X}$, feature $j \in [p]$, radius $\delta > 0$, and model behaviour $g : [x_j - \delta, x_j + \delta] \rightarrow \mathcal{Y}$, we say that the present assumption holds if for the neighbourhood $\mathcal{B} = [x_j - \delta, x_j + \delta] \times \mathcal{X}_{[p] \setminus \{j\}}$,

$$\left\{ f : \forall x' \in \mathcal{B} \ f(x') = g(x'_j), \ f|_{\mathcal{B}^c} \in \mathcal{L}^2|_{\mathcal{B}^c} \right\} \subseteq \mathcal{F}.$$

If the model behaviour g is piecewise linear, then Assumption 2 is satisfied by any $\mathcal{F}$ expressive enough to realize piecewise linear functions. If the user is interested in model behaviour that is more complex than piecewise linear, then it is reasonable to assume that $\mathcal{F}$ is sufficiently expressive to represent this more complex model behaviour locally, and thus Assumption 2 is again satisfied if the model class includes piecewise linear functions away from the region of interest.

3.2 Main Result

Theorem 3.3. Fix any example $x \in \mathcal{X}$, feature $j \in [p]$, radius $\delta > 0$, baseline $\mu \in \mathcal{P}(\mathcal{X})$, and model behaviour $g^{(0)}, g^{(1)} : [x_j - \delta, x_j + \delta] \rightarrow \mathbb{R}$. Suppose that Assumptions 1 and 2 are

satisfied. Let

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \forall x'_j \in [x_j - \delta, x_j + \delta], f(x_{[p] \setminus \{j\}}, x'_j) = g^{(0)}(x'_j) \right\} \\ \mathcal{F}^{(1)} &= \left\{ f \in \mathcal{F} : \forall x'_j \in [x_j - \delta, x_j + \delta], f(x_{[p] \setminus \{j\}}, x'_j) = g^{(1)}(x'_j) \right\}.\end{aligned}$$

For any complete and linear feature attribution method Φ and $\mathbf{h} : \mathbb{R}^{p \times q} \rightarrow [0, 1]$,

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

To interpret this result, consider that for any $\tau \in [0, 1]$, the trivial hypothesis test $\mathbf{h} \equiv \tau$ that ignores the model will achieve

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) = 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

We refer to this family of trivial tests as *random guessing*. For example, one can always achieve $\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) = 0$ at the expense of $\text{Sens}_{\Phi, \mu, x}(\mathbf{h}) = 1$ (by always rejecting the null hypothesis) or $\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) = \text{Sens}_{\Phi, \mu, x}(\mathbf{h}) = 0.5$ (by ignoring the data and using a coin flip). Theorem 3.3 shows that the best tradeoff between sensitivity and specificity that can be achieved by complete and linear feature attribution methods is no better than the tradeoff achieved by random guessing. In other words, this result says that without imposing additional assumptions on the underlying data or learning algorithm to significantly reduce the model complexity, the user cannot conclude that they have learned any information about the model. In particular, they may do no better than random guessing at end-tasks such as recourse and spurious feature identification. In Section 4, we demonstrate that this holds empirically for real data and real models.

For simplicity, we state our results for counterfactual model behaviour with respect to how the model depends on a single feature at a time, and defer the extension to arbitrary groups of features as well as all proofs to Appendix B. The generality of Theorem 3.3 means that it applies to inferring any form of counterfactual model behaviour. Returning to the clinical trial setting of Liu et al. [22] in the introduction, the user may wish to infer if the model output changes as a function of a certain feature (e.g., is the hazard ratio sensitive to the exclusion criteria). To answer this, they must be able to distinguish between $g^{(0)} \equiv 0$ and some $g^{(1)}$ that changes with feature j . However, Theorem 3.3 implies that for every such $g^{(1)}$, the user cannot conclude that they do better than random guessing at this inference task, and therefore, at identifying whether the model depends on the feature. Thus, if this method is used for purposes such as determining membership in a clinical trial, the conclusions are not guaranteed to be any more reliable than random guessing. This has significant implications for using such methods in safety-critical or high stakes applications.

3.3 Proof Sketch of Theorem 3.3

The primary technical result (proved in Appendix B) that facilitates our results is:

Theorem 3.4. *Fix any example $x \in \mathcal{X}$, feature $j \in [p]$, radius $\delta > 0$, baseline $\mu \in \mathcal{P}(\mathcal{X})$, and model behaviour $g : [x_j - \delta, x_j + \delta] \rightarrow \mathcal{Y}$. Suppose that Assumptions 1 and 2 are satisfied. For every attribution $\phi \in \mathbb{R}^q$, there exists a model $f \in \mathcal{F}$ such that for every complete and linear feature attribution method, $\Phi(f, \mu, x)_j = \phi$, and if $|x_j - x'_j| \leq \delta_j$ then $f(x') = g(x'_j)$.*

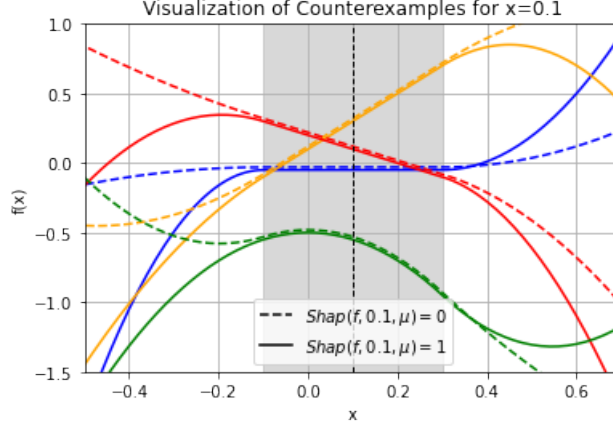


Figure 2: Each line represents a different one-dimensional model. For $x = 0.1$ and $\mu = \text{Unif}(-1, 1)$, dashed lines receive $\text{SHAP}(f, x, \mu) = 0$ while solid lines receive $\text{SHAP}(f, x, \mu) = 1$. The behaviour of models with the same colour is identical within the shaded region, which denotes the neighbourhood $(x - \delta, x + \delta)$ for $\delta = 0.2$. Models can behave very differently and all receive the same attribution (e.g., all dashed lines) and models can be identical in a neighbourhood yet receive very different attribution within that neighbourhood (e.g., lines with the same colour).

Equipped with Theorem 3.4, we use it to prove Theorem 3.3 by constructing a counterexample. For any null hypothesis $\mathcal{F}^{(0)}$ and alternate hypothesis $\mathcal{F}^{(1)}$, we choose $g^{(0)}$ and $g^{(1)}$ satisfying the counterfactual model behaviour prescribed by the null hypothesis and alternate hypothesis respectively. For example, in the recourse setting, $g^{(0)}$ may be increasing in the feature while $g^{(1)}$ is decreasing. Then, by Theorem 3.4, we can construct $f^{(0)}$ and $f^{(1)}$ that locally agree with $g^{(0)}$ and $g^{(1)}$ respectively, yet receive the same feature attribution. Consequently, any feature-attribution hypothesis test that relies on only the feature attribution to draw conclusions cannot distinguish between $f^{(0)}$ and $f^{(1)}$, and hence has been provided no additional information to do better than random guessing. While a single counterexample is sufficient to prove Theorem 3.3, the proof of Theorem 3.4 reveals that we can actually find uncountably many models that work as counterexamples, which has important implications for practical settings. We demonstrate this in our on-average results (Section 3.5) and experiments (Section 4).

We visualize the intuition behind this result in Figure 2, where we show (a) models can behave very differently and all receive the same attribution and (b) models can be identical in a local neighbourhood of interest yet receive very different attribution. Critically, our counterexamples are very simple (piecewise linear in the proof, piecewise quadratic for smooth visualization in Figure 2), and hence easily realized by neural networks. Finally, since feature attribution methods provide only a summary of the model, it is clear that one can always find *some* example of two different models that receive the same attribution. The primary insight that enables our results is that we can always find models that *differ in exactly the way that we wish to test* yet receive the same attribution. The next natural question is: how does this result extend to end-tasks that practitioners care about?

3.4 Results for End-Tasks of Interest

We discuss three end-tasks, starting with the simplest one (local model behaviour) which forms the basis of the two more common end-tasks: recourse and spurious correlation. There are multiple notions of these tasks in the literature, and they are often used without a precise formalization. To facilitate analysis, we provide *one possible* formalization of the easiest version of these tasks. We defer general definitions and full proofs to Appendix C.

3.4.1 Simple Local Model Behaviour

A consequence of Theorem 3.3 is that if the notion of counterfactual model behaviour is sufficiently local, SHAP and IG may not provide valid inference for f . Specifically, we consider the task of identifying whether a model output is locally sensitive to perturbations of a feature, which is a necessary component of identifying spurious features: if one can reliably identify spurious features, one must be able to identify features for the which the model output is insensitive to small perturbations. We first show that Gradient is *always* successful at this task for a sufficiently small notion of perturbations. While the limitations of Gradient are well-known for many tasks [3] and there are additional challenges for Gradient computed using backpropagation [29], the following result demonstrates that there exists a task for which Gradient is reliable yet SHAP and IG may not be.

Proposition 3.5. *Fix $x \in \mathcal{X}$ and suppose $\mathcal{F} \subseteq \{f : \mathbb{R}^p \rightarrow \mathbb{R}, \nabla f(x) \text{ exists}\}$. For every $\varepsilon > 0$, $x \in \mathcal{X}$, and $j \in [p]$, there exists $\delta > 0$ such that if*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{f \in \mathcal{F} : \sup_{\alpha \leq \delta} |f(x + \alpha e_j) - f(x)| \leq \delta\varepsilon/2\right\} \\ \mathcal{F}^{(1)} &= \left\{f \in \mathcal{F} : \sup_{\alpha \leq \delta} |f(x + \alpha e_j) - f(x)| > \delta\varepsilon\right\},\end{aligned}$$

then there exists $\mathbf{h}$ using $\Phi = \text{Gradient}$ such that for every $\mu \in \mathcal{P}(\mathcal{X})$

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) = \text{Sens}_{\Phi, \mu, x}(\mathbf{h}) = 1.$$

Using Theorem 3.3, we next show that complete and linear feature attribution methods provably are less reliable than Gradient for this task.

Proposition 3.6. *Fix $x \in \mathcal{X}$ and suppose $\{f : \mathbb{R}^p \rightarrow \mathbb{R}, f \text{ is piecewise linear and } \nabla f(x) \text{ exists}\} \subseteq \mathcal{F} \subseteq \{f : \mathbb{R}^p \rightarrow \mathbb{R}, \nabla f(x) \text{ exists}\}$. For every sufficiently small $\varepsilon, \delta > 0$, $x \in \mathcal{X}$, and $j \in [p]$, if*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{f \in \mathcal{F} : \sup_{\alpha \leq \delta} |f(x + \alpha e_j) - f(x)| \leq \delta\varepsilon/2\right\} \\ \mathcal{F}^{(1)} &= \left\{f \in \mathcal{F} : \sup_{\alpha \leq \delta} |f(x + \alpha e_j) - f(x)| > \delta\varepsilon\right\},\end{aligned}$$

then for every feature-attribution hypothesis test $\mathbf{h}$, complete and linear feature attribution method Φ , and $\mu \in \mathcal{P}(\mathcal{X})$ satisfying Assumption 1,

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

3.4.2 Recourse and Spurious Features

While Section 3.4.1 proves that complete and linear feature attribution methods can be unreliable for inferring sufficiently local counterfactual model behaviour, common end-tasks in the literature are only “moderately” local. We now show that inference for these tasks can also be as uninformative as random guessing. To do so, we formalize increasing and decreasing a feature by considering a distribution over perturbations and measuring the model change on average with respect to this distribution (which may differ from the baseline used to compute the feature attribution).

Definition 3.7 (Recourse). *Fix any $x \in \mathcal{X}$, $j \in [p]$, $k \in [q]$, $\delta > 0$, and $\nu \in \mathcal{P}(\mathcal{X})$. Let*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \mathbb{E}_{X \sim \nu} [f(x_{[p] \setminus \{j\}}, X_j)_k \mid X_j \in [x_j, x_j + \delta]] \right. \\ &\quad \left. > \mathbb{E}_{X \sim \nu} [f(x_{[p] \setminus \{j\}}, X_j)_k \mid X_j \in [x_j - \delta, x_j]] \right\} \\ \mathcal{F}^{(1)} &= \mathcal{F} \setminus \mathcal{F}^{(0)}.\end{aligned}$$

Recourse (Informal). Here, ν is a distribution from which perturbed examples can be sampled from. This distribution need not be the same as the baseline used by the feature attribution method—a common choice would be the uniform distribution or a pointmass for a single perturbation of interest. Then, $\mathcal{F}^{(0)}$ corresponds to those models where increasing feature j will increase the model output on average. Again as a concrete example, if the model output is the probability of loan acceptance and feature j is income, the user asks whether to increase or decrease credit score in order to improve the probability of acceptance. Next, we formalize distinguishing whether a model is sensitive or insensitive to perturbations.

Definition 3.8 (Spurious Features). *Fix any $x \in \mathcal{X}$, $j \in [p]$, $k \in [q]$, $\delta > 0$, and $\varepsilon > 0$. Let*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \sup_{x'_j \in (x_j, x_j + \delta]} |f(x_{[p] \setminus \{j\}}, x'_j)_k| = 0 \right\} \\ \mathcal{F}^{(1)} &= \left\{ f \in \mathcal{F} : \sup_{x'_j \in (x_j, x_j + \delta]} |f(x_{[p] \setminus \{j\}}, x'_j)_k| \geq \varepsilon \right\}.\end{aligned}$$

Spurious Features (Informal). $\mathcal{F}^{(0)}$ corresponds to those models where the model output does not change from perturbing feature j . Meanwhile, $\mathcal{F}^{(1)}$ corresponds to the models that have a “significant” change from perturbing feature j , where significance is encoded by the size of ε . Concretely, if features correspond to the pixels of a watermark on an X-ray, the user asks whether the model output is sensitive to the values of these pixels.

We now apply Theorem 3.3 to these definitions and obtain the following implications.

Corollary 3.9. *Fix any $x \in \mathcal{X}$, $j \in [p]$, $\delta > 0$, and $\mu \in \mathcal{P}(\mathcal{X})$ such that Assumption 1 is satisfied. Fix $k \in [q]$ and $\nu \in \mathcal{P}(\mathcal{X})$, and let $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$ be as defined in either Definition 3.7 or Definition 3.8. Suppose that there exists $f^{(0)} \in \mathcal{F}^{(0)}$ and $f^{(1)} \in \mathcal{F}^{(1)}$ such that $g^{(0)} = f^{(0)}|_{\mathcal{B}}$ and $g^{(1)} = f^{(1)}|_{\mathcal{B}}$ each satisfy Assumption 2. Then for any complete and linear feature attribution method Φ and feature-attribution hypothesis test $\mathbf{h}$,*

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

As previously mentioned, Corollary 3.9 implies that the user cannot distinguish whether increasing or decreasing the feature is the correct direction to increase the model prediction. For recourse, the main assumption that Assumption 2 holds is satisfied in this case when $\mathcal{F}$ contains piecewise linear functions, since the functions $g(x_j) = x_j$ for $f^{(0)}$ and $g(x_j) = -x_j$ for $f^{(1)}$ suffice. Similarly, Corollary 3.9 implies that the user cannot distinguish whether the model prediction is sensitive to changes in the feature. For spurious features, the main assumption that Assumption 2 holds is again satisfied when $\mathcal{F}$ contains piecewise linear functions, since the functions $g(x_j) = 0$ for $f^{(0)}$ and $g(x_j) = \varepsilon$ for $f^{(1)}$ suffice.

3.5 Average Local Performance for Simple Models

As motivated in Section 2, the usual definitions of specificity and sensitivity demand accurate hypothesis tests for every model of interest. We now show that for standard distributions placed over simple model classes, our results apply even for the *easier* task of obtaining accurate hypothesis tests on average over the models.

Consider the simple class of univariate models $\mathcal{F} = \{x \mapsto ax^n - x : a \in \mathbb{R}\}$ for some fixed $n \geq 2$. Although this model class is so simple that Assumption 2 does not apply (and hence we cannot directly apply Theorem 3.3), we can prove a similar result that applies on average.

Proposition 3.10. *Let π denote the distribution over $\mathcal{F}$ induced by $a \sim \text{Gaussian}(0, 1)$, let μ be such that $\mathbb{E}_\mu X^n \in (1/2, 1)$, and set $x = 1$. Then, for any complete and linear Φ ,*

$$\mathbb{P}_{f \sim \pi} [\text{sgn}(\Phi(f, \mu, x)) \neq \text{sgn}(f'(x))] \in (0.25, 0.5).$$

As a consequence, any feature-attribution hypothesis test that relies on the sign of a complete and linear feature attribution method to infer local counterfactual model behaviour will draw the wrong conclusion at least 1/4 of the time, even for this exceedingly simple model class. In the next section, we show empirically that our results apply on average in the more complex settings for which Theorem 3.3 applies.

4 Experiments

The theoretical guarantees of the previous section demonstrate that common feature attribution methods such as SHAP and IG cannot reliably infer counterfactual model behaviour. While our assumptions are satisfied by moderately rich model classes (including neural networks), our theory does not rule out the possibility that additional structure extracted from the training data or learning algorithm is further aiding feature attribution methods in practice. In this section, we provide experimental results consistent with what our theory predicts, even when we restrict consideration to neural networks trained with stochastic gradient descent on real datasets. Specifically, for ReLU neural networks on tabular data and convolutional neural networks on image data, we observe that SHAP and IG are close to random guessing for the end-tasks of algorithmic recourse and spurious feature identification. We also compare with three common local methods: gradients, SmoothGrad, and LIME. We find that for simple tabular data, these methods can outperform SHAP and IG, while for image data all methods are comparable to random guessing. All code is available at <https://github.com/google-research/interpretability-theory>.

4.1 Methods

To visualize specificity and sensitivity, we use the standard receiver operating characteristic (ROC) curve, which shows the trade-off of the false positive rate ($1 - \text{specificity}$) on the x -axis and the true positive rate (sensitivity) on the y -axis. An ROC curve is computed by varying the rejection threshold for a hypothesis test: the more strict the threshold, the less likely to reject the null, and hence the lower the false positive rate. An ideal hypothesis test threshold achieves the top left corner of the plot (0% false positives and 100% true positives), and generally a hypothesis test is better if the curve is closer to the top left corner of the plot. The diagonal line from $(0, 0)$ to $(1, 1)$ is the line $\text{specificity} = 1 - \text{sensitivity}$, and hence corresponds exactly to random guessing.

For each dataset, feature attribution method, and end-task, we construct an empirical ROC curve. To do so, for each dataset we retrain 10 neural networks (models) using different random seeds to similar accuracy. Then, we randomly sample 20 examples from the test dataset and compute each feature attribution method on each example for each model. For each non-categorical feature and each end-task, we compute a hypothesis test at a specific threshold using the feature attribution and compare it to a “ground truth”; for details on the hypothesis tests and the ground truth, see Section 4.1.1. Thus, each point on a plot corresponds to a single dataset, model, feature attribution method, end-task, and hypothesis test threshold, and represents the empirical true and false positive rates for a hypothesis test averaged over all features on 20 examples. We repeat each of these calculations with 40 different hypothesis test thresholds to create the entire ROC curve. Since we reuse the same 20 examples for each model, the noisy ROC curve is actually comprised of 10 different (one for each model) monotonic ROC curves.

For image data, it is ambiguous what constitutes a “feature”. Following the literature [26], we consider each individual pixel to be a feature. For computational reasons, we did not average over every pixel for each hypothesis test, but instead over a sample of 10 pixels (this matches that most of the tabular datasets have roughly 10 features).

4.1.1 End-Tasks

Ground Truth. For both end-tasks, the ground truth relies on a neighbourhood around examples. For each example x and feature j we consider, we construct this neighbourhood as follows. We fix a percentage $p \in [0, 1]$, and compute a range R which is the maximum value of the j th feature minus the minimum value of the j th feature on the dataset under consideration. We then create 20 copies of x , where all feature values are fixed except for the j th, which is set to $x'_j = x_j + \delta$ for δ evenly spaced in $(-pR, pR)$. We display results for $p = 0.1$, but found similar results for p ranging from 0.5 to 0.01. For smaller p , we encountered floating point issues (the model output didn’t change at all over the neighbourhood).

Recourse. We use the task of Definition 3.7 for a single feature, with ν taken to be the uniform distribution. To compute the ground truth, we approximate expectation under this distribution by comparing the empirical average model output on the first half of the perturbed examples to the empirical average model output on the second half of the perturbed examples. That is, the ground truth is 1 if the average model output is larger from increasing the feature versus decreasing it, and 0 otherwise.

To conduct the hypothesis test, we use the sign and magnitude of the feature attribution. In particular, for threshold $\alpha \in \mathbb{R}$ and feature attribution $\phi \in \mathbb{R}$, we use $\mathbf{h}(\phi) = \mathbb{I}\{\phi > \alpha\}$. This hypothesis test is consistent with applications in the literature: Jain et al. [14] use SHAP (also accounting for magnitude) and Ghosh et al. [12] use IG to identify which features should be adjusted for to achieve outcome fairness.

Spurious Features. We use a variant of the task of Definition 3.8, replacing sup with variance for better stability. First, we compute the perturbation described above for 100 additional examples from the dataset of interest and then compute the variance of the model output over each perturbation (providing an empirical distribution of such variances). We then set the ε in Definition 3.8 to be the 80th quantile of this empirical distribution, and the ground truth is computed by comparing the model output variance over the perturbations of the example at hand with this ε . That is, the ground truth is 1 if perturbing the feature causes the model output to vary more than 80% of other features, and 0 otherwise.

To conduct the hypothesis test, we use only the magnitude of the feature attribution. In particular, for threshold $\alpha \in \mathbb{R}$ and feature attribution $\phi \in \mathbb{R}$, we use $\mathbf{h}(\phi) = \mathbb{I}\{|\phi| > \alpha\}$.

4.1.2 Feature Attribution Methods

SHAP. For tabular data (i.e., a small number of features), we compute SHAP according to Definition A.1 using the `KernelExplainer` function from the Python `shap` package [23]. We approximate the outer expectation (with respect to the training data distribution) using an empirical average over 100 samples and the inner expectation (with respect to the Shapley kernel over subsets) using an empirical average over 500 samples. For image data, the number of subsets of pixels is too large to approximate accurately, so we follow the `shap` documentation and use the `Explainer` function with `maskers.Image`. This approximates the SHAP value by “blocking out” groups of pixels.

Gradient. We compute ∇f exactly using TensorFlow [2].

Integrated Gradients. We use Definition A.2, approximating the integral using a sum with 20 steps. Following Sundararajan et al. [41], we use a pointmass baseline, so the expectation does not need to be approximated. We visualize two baselines: all features set to zero (the mean of the data after rescaling), and all features set to their minimum value.

SmoothGrad. We use Definition A.7, where for each x we take μ to be $\text{Gaussian}(x, 0.1 \cdot I_p)$ following Smilkov et al. [37]. We approximate this expectation using 100 samples.

LIME. We use Definition A.8, following the use of the best (regularized) local linear model from Ribeiro et al. [33]. For simplicity, we set $\lambda = 1$ and for each x take μ to be $\text{Gaussian}(x, 0.1 \cdot I_p)$ —in Ribeiro et al. [33], μ is denoted by π_x . We then approximate the expectation using 100 samples so that the arg min can be found exactly using the closed-form solution of regularized least squares.

4.1.3 Datasets

Tabular data. We consider 5 standard tabular datasets from the UCI repository [8]: wine origin (`wine`) [11], credit approval (`credit`) [32], chess outcome (`chess`) [6], E. coli protein localization (`ecoli`) [27], and abalone age (`abalone`) [28]. After centering and normalizing

the features by their empirical standard deviation, we trained small ReLU neural networks on the first four datasets to average test accuracies of 100%, 80%, 99%, and 85% respectively, while for the `abalone` dataset (which has integer responses) we achieved average test mean squared error of 4.5.

Image data. We consider 3 standard image datasets: MNIST digit classification (`mnist`) [20], Fashion-MNIST classification (`fashion`) [45], and CIFAR-10 image classification (`cifar-10`) [18]. After normalizing the pixel values to $[0, 1]$, we trained standard convolutional neural networks to average test accuracies of 99%, 90%, and 80% respectively.

4.2 Results

We plot ROC curves for all end-tasks, datasets, and attribution methods in Figure 3 (tabular datasets) and Figure 4 (image datasets). We highlight some observations:

Observation 1: SHAP and Integrated Gradients ROC curves are near random guessing for almost all experiments. This agrees with what our theory suggests.

Observation 2: The baseline for IG matters. Using a baseline of all zeroes is, in general, worse than using a baseline corresponding to the minimum features. This difference is to be expected: the choice of baseline is already empirically known to heavily influence the output [39]. However, we emphasize that (a) both baselines are often near the random guessing line, and (b) *without strong additional knowledge about the dataset and learned model, there is no way to know in advance what the “right” baseline is to choose for your task.*

Observation 3: Simpler, local feature attribution methods sometimes outperform SHAP and IG. For some tasks, gradients, SmoothGrad, and LIME perform much better than random guessing, likely because algorithmic recourse and spurious feature identification are local end-tasks. These methods are not foolproof, however; for many cases they also fail to improve on random guessing. Once again, *there is no way to know in advance whether your feature attribution method will work for your task without strong additional knowledge.*

Observation 4: The ROC curves suggest that, in general, the end-tasks are easier to solve on tabular datasets than image datasets. While for tabular datasets we see many ROC curves far from random guessing, the ROC curves for image datasets are near the diagonal that corresponds to random guessing. We conjecture that this is because the models learned for tabular datasets can be quite simple, making gradients (and their variants) more indicative of counterfactual model behaviour. (Recall that if the model is linear, counterfactual model behaviour is recovered exactly by all methods.)

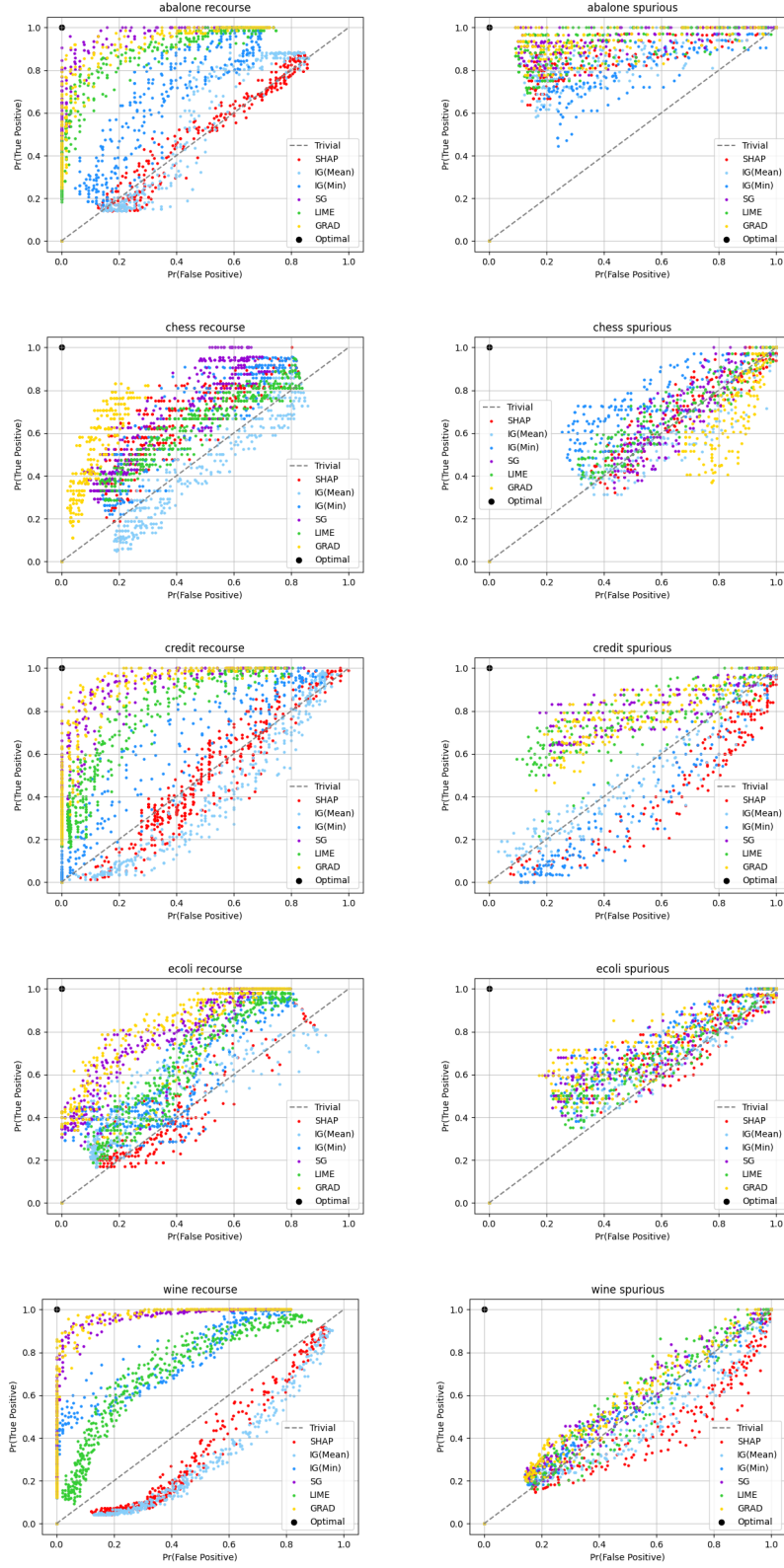


Figure 3: Visualizing ROC curves for tabular datasets. A feature attribution method is better for an end-task if the ROC curve is closer to the top left corner on average.

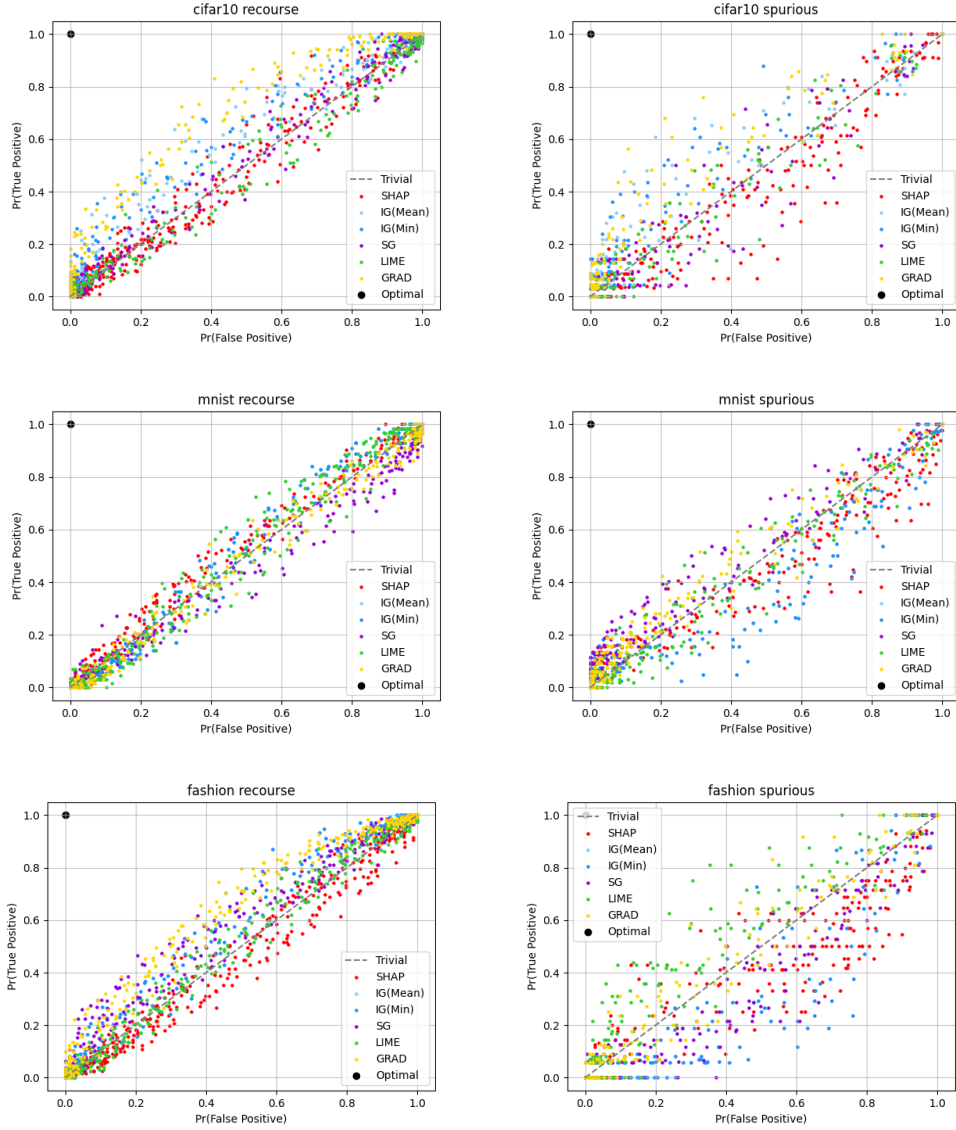


Figure 4: Visualizing ROC curves for image datasets. A feature attribution method is better for an end-task if the ROC curve is closer to the top left corner on average.

5 Towards Theoretical Guarantees for Perturbation-Based Methods

We have shown both theoretically and empirically that complete and linear feature attribution methods like SHAP and IG can be unreliable for solving end-tasks like algorithmic recourse and spurious feature identification. At the same time, while simpler feature attribution methods like Gradient can be accurate for inferring sufficiently local model behaviour, this can also clearly fail even for simple tasks that demand understanding a moderately local region. So, faced with solving a given end-task, what should a practitioner do?

To answer this question, we turn to the existing literature on feature attribution methods that are empirically better than SHAP and IG for many tasks. For example, Fong and Vedaldi [10] learn a blurring mask that changes the class probability while simultaneously maximizing how “informative” the mask is, Petsiuk et al. [30] use randomly generated masks to learn which perturbations maximally change the model output, Kapishnikov et al. [16] use image masking to find regions that maximally change the class probability, Qi et al. [31] and Khorram et al. [17] learn a feature mask based on both removing and adding components to the image to change the class probability, and Shitole et al. [35] combine multiple saliency maps into a single graph to illustrate multiple different minimal perturbations to change the model output.

However, despite the many methods listed above, mathematically establishing when these methods are (and are not) reliable has remained elusive. As a first step towards such theory, we consider a method that is *guaranteed* to work: brute-force solving end-tasks via repeated model evaluations. In particular, since counterfactual model behaviour is determined by $f(x')$ for x' near x , this can always be inferred by computing $f(x')$ (or, perhaps, $\nabla f(x')$) at sufficiently many examples x' . To demonstrate this, we state the following simple result showing how spurious feature identification (i.e., Definition 3.8) relies on the number of queries. To avoid additional notational burden, we state this result informally, and defer precise notation, more general statements, and proofs to Appendix D.

Theorem 5.1. *Suppose that $\mathcal{Y} = \mathbb{R}$ and there exists $L > 0$ such that all $f \in \mathcal{F}$ are L -Lipschitz. For fixed $\delta, \varepsilon > 0$ and $j \in [p]$, consider the end-task of Definition 3.8 with $x = 0$. For every $n \in \mathbb{N}$, there exists a hypothesis test $\mathbf{h}^{(n)}$ that uses only n evaluations of f yet*

$$\text{Spec}(\mathbf{h}^{(n)}) = 1 \quad \text{and} \quad \text{Sens}(\mathbf{h}^{(n)}) = 1 - \left(1 - \frac{2\varepsilon}{L\delta}\right)^n.$$

A few remarks on Theorem 5.1. First, this result implies that the user can achieve specificity *and* sensitivity arbitrarily close to one by evaluating f sufficiently many times (i.e., taking n to infinity). Second, while we did not state this in Section 3 for simplicity, from the proof of Theorem 3.3 it can be observed that Assumption 2 can be relaxed to require only Lipschitz piecewise linear functions, and hence this additional structure is not sufficient to circumvent our impossibility result for complete and linear feature attribution methods (i.e., Corollary 3.9). Third, we defer the precise definition of the hypothesis test to Appendix D, but here is a simple intuition: uniformly sample n examples $x^{(1:n)}$ and reject the null hypothesis if and only if $f(0, x_j^{(t)}) > \varepsilon$ for at least one t . Finally, in Appendix D we

extend this result to the multivariate case and quantify the corresponding dependence on p (unsurprisingly, this dependence is exponential), and we show that this simple hypothesis test achieves nearly optimal worst-case dependence on parameters like δ , ε , L , p , and n .

As a concrete example, consider the spurious feature identification end-task for 10 features (say, 10 pixels where a watermark appears) on 1-Lipschitz models. To identify if the model is sensitive (model output changes by more than 1%) to a 5% change in the feature value, we show it is possible to get perfect specificity and over 90% sensitivity with roughly 20,000 model evaluations.

While the end-task of spurious feature identification can be solved by brute-force evaluations (and a similar argument can be made for other counterfactual model behaviour end-tasks), there is much work to be done. In particular, the simple hypothesis test in Theorem 5.1 is designed to always succeed, but may be quite inefficient for more structured models. Unfortunately, for the existing brute-force hypothesis tests that are more efficient, it is unclear for which end-tasks they provably work. Our hypothesis testing framework allows us to rigorously evaluate such methods, and our formalization of end-tasks suggests that techniques and methods from optimization theory may be a useful starting point.

6 Related Literature

While theory is sparse for feature attribution, some other impossibility results have recently appeared in the literature.

Srinivas and Fleuret [38] prove that any complete feature attribution method cannot be “weakly dependent” on the input. Intuitively, weak dependence is a type of stability, and is precisely defined as the feature attribution method output depending only on the weights if the model is piecewise linear. One implication of our results is that any complete and linear feature attribution method cannot even be *close* to weakly dependent, since this would allow a hypothesis test to extract useful information about the weights for piecewise linear models (and hence beat random guessing at tasks like recourse or spurious features).

Fokkema et al. [9] prove that continuous feature attribution methods (including SHAP and IG) can fail for algorithmic recourse at examples near the decision boundary of a model (i.e., where the true recourse direction should switch). In contrast, we prove that for sufficiently rich model classes, recourse-like counterfactual model behaviour cannot be reliably inferred at *any* example. Moreover, we prove that these methods fail to infer general counterfactual model behaviour, with recourse impossibility being a corollary of our results.

Han et al. [13] show that for perturbation-based feature attribution methods (including SHAP and IG), there will always exist a neighbourhood where the feature attribution does not match the model (in the sense of local function approximation). In contrast, our results imply that this holds for exactly the neighbourhood where the feature attribution method is centered; that is, not only do feature attribution methods fail *somewhere* to capture model behaviour globally, they fail to do so *in the exact local region of interest* as well.

Beyond such impossibility results, others have proposed various ways to formalize the task of feature attribution. Watson et al. [44] use the concepts of necessity and sufficiency to discuss whether feature attributions can be tied to model behaviour. Watson and Floridi [43] propose

studying interpretability as a decision theory problem, where two players iterate to find a “good” explanation, which they formalize as learning a local approximation of the model. Afchar et al. [4] formalize ground truth for feature attribution methods using a local form of functional feature dependence. Zhou and Shah [50] formalize feature attribution methods as loss minimizers, which they use to unify certain methods. While all of these formalizations provide a useful lens for studying feature attribution methods, none have been used to prove results about their performance. Our hypothesis testing framework formalizes (some of) these ideas and enables concrete mathematical reasoning about the performance of feature attribution methods.

Finally, there exist many individual counterexamples for common feature attribution methods in the literature. Sundararajan and Najmi [40] unify various versions of Shapley value methods and provide an example where SHAP and IG differ, for which the authors point out that it is not clear which one is correct. Kumar et al. [19] provide simple counterexamples where SHAP can be computed analytically yet result in attributions that disagree with human intuition. Merrick and Taly [25] and Janzing et al. [15] show that the conditional version of SHAP can give large attribution to completely irrelevant features. Our results are consistent with the above findings, providing a way to formally compare feature attribution methods (hypothesis testing) along with general, mild conditions under which common methods can have poor performance. Characterizing other feature attribution methods in a similar way and refining the assumptions that lead to performance guarantees is critical to better understanding the utility of feature attribution as part of a user’s toolkit.

7 Conclusion

The current deployment of feature attribution methods in high-stakes settings such as medicine and law demands a rigorous understanding of their performance. In this work, we characterized conditions under which common feature attribution methods provably are unreliable when used to infer behaviour for learned models. Our work concludes that using feature attribution as is currently prescribed in the literature need not improve on random guessing at inferring model behavior. The implications of our work are two-fold. First, it is necessary to build more structure into these methods in order to improve upon the current performance guarantees. We show that a simple brute-force method (easy to define but computationally expensive) is guaranteed to work for *some end-tasks*. As a result, our second implication is that defining the end-task accurately and concretely is crucial. Once given an end-task, seeking a method that directly optimizes the task can provide users with straightforward answers. In summary, the goals of interpretability are not impossible to achieve, but they may require new methods along with a precise understanding of the assumptions under which such methods are reliable. The development of methods that come equipped with performance guarantees and enjoy computational efficiency remains a major open challenge for future work.

Acknowledgements

BB acknowledges support from the Vector Institute. We thank Astrid Bertrand, Robert Geirhos, Adam Pearce, Lisa Schut, Martin Wattenberg, and Qiqi Yan for helpful feedback on preliminary drafts.

References

- [1] Kjersti Aas, Martin Jullum, and Anders Løland. Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. *Artificial Intelligence*, 298:103502–103526, 2021.
- [2] Martín Abadi, Ashish Agarwal, Paul Barham, Eugene Brevdo, Zhifeng Chen, Craig Citro, Greg S. Corrado, Andy Davis, Jeffrey Dean, Matthieu Devin, Sanjay Ghemawat, Ian Goodfellow, Andrew Harp, Geoffrey Irving, Michael Isard, Yangqing Jia, Rafal Jozefowicz, Lukasz Kaiser, Manjunath Kudlur, Josh Levenberg, Dandelion Mané, Rajat Monga, Sherry Moore, Derek Murray, Chris Olah, Mike Schuster, Jonathon Shlens, Benoit Steiner, Ilya Sutskever, Kunal Talwar, Paul Tucker, Vincent Vanhoucke, Vijay Vasudevan, Fernanda Viégas, Oriol Vinyals, Pete Warden, Martin Wattenberg, Martin Wicke, Yuan Yu, and Xiaoqiang Zheng. TensorFlow: Large-scale machine learning on heterogeneous systems, 2015. URL <https://www.tensorflow.org/>. Software available from tensorflow.org.
- [3] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems 32*, 2018.
- [4] Darius Afchar, Romain Hennequin, and Vincent Guigue. Towards rigorous interpretations: A formalisation of feature attribution. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.
- [5] Raman Arora, Amitabh Basu, Poorya Mianjy, and Anirbit Mukherjee. Understanding deep neural networks with rectified linear units. In *Proceedings of the 6th International Conference on Learning Representations*, 2018.
- [6] Michael Bain and Stephen Muggleton. Learning optimal chess strategies. In *Machine Intelligence 13: Machine Intelligence and Inductive Learning*, pages 291–309. 1994.
- [7] Kuan-Lin Chen, Harinath Garudadri, and Bhaskar D. Rao. Improved bounds on neural complexity for representing piecewise linear functions. In *Advances in Neural Information Processing Systems 36*, 2022.
- [8] Dheeru Dua and Casey Graff. UCI machine learning repository, 2017. URL <http://archive.ics.uci.edu/ml>.
- [9] Hidde Fokkema, Rianne de Heide, and Tim van Erven. Attribution-based explanations that provide recourse cannot be robust, 2022. arXiv:2205.15834.

- [10] Ruth C Fong and Andrea Vedaldi. Interpretable explanations of black boxes by meaningful perturbation. In *Proceedings of the 2017 IEEE International Conference on Computer Vision*, 2017.
- [11] M. Forina, C. Armanino, M. Castino, and M. Ubigli. Multivariate data analysis as a discriminating method of the origin of wines. *Journal of Grapevine Research*, 25(3), 1986.
- [12] Avijit Ghosh, Aalok Shanbhag, and Christo Wilson. Faircanary: Rapid continuous explainable fairness. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, 2022.
- [13] Tessa Han, Suraj Srinivas, and Himabindu Lakkaraju. Which explanation should I choose? A function approximation perspective to characterizing post hoc explanations. In *Advances in Neural Information Processing Systems 36*, 2022.
- [14] Aditya Jain, Manish Ravula, and Joydeep Ghosh. Biased models have biased explanations, 2020. arXiv:2012.10986.
- [15] Dominik Janzing, Lenon Minorics, and Patrick Blöbaum. Feature relevance quantification in explainable AI: A causal problem. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, 2020.
- [16] Andrei Kapishnikov, Tolga Bolukbasi, Fernanda Viegas, and Michael Terry. XRAI: Better attributions through regions. In *2019 IEEE/CVF International Conference on Computer Vision*, 2019.
- [17] Saeed Khorram, Tyler Lawson, and Li Fuxin. iGOS++ integrated gradient optimized saliency by bilateral perturbations. In *Proceedings of the 2021 Conference on Health, Inference, and Learning*, 2021.
- [18] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, 2009.
- [19] I. Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle A. Friedler. Problems with Shapley-value-based explanations as feature importance measures. In *Proceedings of the 37th International Conference on Machine Learning*, 2020.
- [20] Yann LeCun, Corinna Cortes, and CJ Burges. MNIST handwritten digit database. *ATT Labs [Online]*. Available: <http://yann.lecun.com/exdb/mnist>, 2, 2010.
- [21] Miguel Lerma and Mirtha Lucas. Symmetry-preserving paths in integrated gradients, 2021. arXiv:2103.13533.
- [22] Ruishan Liu, Shemra Rizzo, Samuel Whipple, Navdeep Pal, Arturo Lopez Pineda, Michael Lu, Brandon Arnieri, Ying Lu, William Capra, Ryan Copping, and James Zou. Evaluating eligibility criteria of oncology trials using real-world data and AI. *Nature*, 592:629–633, 2021.

- [23] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 31*, 2017.
- [24] Kevin McCloskey, Ankur Taly, Frederico Monti, Michael P. Brenner, and Lucy J. Colwell. Using attribution to decode binding mechanism in neural network models for chemistry. *Proceedings of the National Academy of Sciences of the United States of America*, 116(24):11624–11629, 2019.
- [25] Luke Merrick and Ankur Taly. The explanation game: Explaining machine learning models using Shapley values. In *International Cross-Domain Conference for Machine Learning and Knowledge Extraction*, 2020.
- [26] Christoph Molnar. *Interpretable Machine Learning*. 2 edition, 2022. URL <https://christophm.github.io/interpretable-ml-book>.
- [27] Kenta Nakai and Minoru Kanehisa. Expert system for predicting protein localization sites in gram-negative bacteria. *Proteins: Structure, Function, and Bioinformatics*, 11(2):95–110, 1991.
- [28] Warwick J Nash, Tracy L Sellers, Simon R Talbot, Andrew J Cawthorn, and Wes B Ford. The population biology of Abalone (*haliotis* species) in Tasmania. i. Blacklip Abalone (*h. rubra*) from the north coast and islands of Bass Strait. *Sea Fisheries Division, Technical Report*, 48:p411, 1994.
- [29] Weili Nie, Yang Zhang, and Ankit B. Patel. A theoretical explanation for perplexing behaviors of backpropagation-based visualizations. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.
- [30] Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *Proceedings of the 29th British Machine Vision Conference*, 2018.
- [31] Zhongang Qi, Saeed Khorram, and Fuxin Li. Visualizing deep networks by optimizing with integrated gradients. In *Workshop at the 34th IEEE Conference on Computer Vision and Pattern Recognition*, 2019.
- [32] J. Ross Quinlan. Simplifying decision trees. *International journal of man-machine studies*, 27(3):221–234, 1987.
- [33] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “Why should I trust you?” Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2016.
- [34] Joanna Roder, Laura Maguire, Robert Georgantas III, and Heinrich Roder. Explaining multivariate molecular diagnostic tests via Shapley values. *BMC Medical Informatics and Decision Making*, 21(211), 2021.

- [35] Vivswan Shitole, Fuxin Li, Minsuk Kahng, Prasad Tadepalli, and Alan Fern. One explanation is not enough: Structured attention graphs for image classification. *Advances in Neural Information Processing Systems 35*, 2021.
- [36] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps, 2013. arXiv:1312.6034.
- [37] Daniel Smilkov, Nikhil Thorat, Been Kim, Fernanda Viegas, and Martin Wattenberg. SmoothGrad: Removing noise by adding noise. In *Proceedings of the ICML 2017 Workshop on Visualization for Deep Learning*, 2017.
- [38] Suraj Srinivas and François Fleuret. Full-gradient representation for neural network visualization. In *Advances in Neural Information Processing Systems 33*, 2019.
- [39] Pascal Sturmfels, Scott Lundberg, and Su-In Lee. Visualizing the impact of feature attribution baselines. *Distill*, 2020. doi: 10.23915/distill.00022. <https://distill.pub/2020/attribution-baselines>.
- [40] Mukund Sundararajan and Amir Najmi. The many Shapley values for model explanation. In *Proceedings of the 37th International Conference on Machine Learning*, 2020.
- [41] Mukund Sundararajan, Ankur Taly, and Qiyi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.
- [42] Ugo Tanielian, Maxime Sangnier, and Gerard Biau. Approximating Lipschitz continuous functions with GroupSort neural networks. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, 2021.
- [43] David S. Watson and Luciano Floridi. The explanation game: A formal framework for interpretable machine learning. *Synthese*, 198:9211–9242, 2021.
- [44] David S. Watson, Limor Gultchin, Ankur Taly, and Luciano Floridi. Local explanations via necessity and sufficiency: Unifying theory and practice. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, 2021.
- [45] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, 2017. arXiv:1708.07747.
- [46] Chih-Kuan Yeh, Cheng-Yu Hsieh, Arun Sai Suggala, David I. Inouye, and Pradeep Ravikumar. On the (in)fidelity and sensitivity of explanations. In *Advances in Neural Information Processing Systems 33*, 2019.
- [47] Jacob Yerushalmy. Statistical problems in assessing methods of medical diagnosis, with special reference to X-ray techniques. *Public Health Reports (1896-1970)*, 62(40):1432–1449, 1947.

- [48] Mohammad Zaeri-Amirani, Fatemeh Afghah, and Sajad Mousavi. A feature selection method based on Shapley value to false alarm reduction in ICUs, a genetic-algorithm approach. In *Proceedings of the 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2018.
- [49] Kaiyue Zhou, Suzan Arslanturk, Douglas B. Craig, Elisabeth Heath, and Sorin Draghici. Discovery of primary prostate cancer biomarkers using cross cancer learning. *Scientific Reports*, 11(10433), 2021.
- [50] Yilun Zhou and Julie Shah. The solvability of interpretability evaluation metrics, 2022. arXiv:2205.08696.

A Proofs for Completeness and Linearity

We begin with precise definitions of SHAP and Integrated Gradients.

For SHAP, there are two possible definitions in the literature, which respectively use marginal and conditional expectations. In practice, the most common implementation of SHAP uses the marginal definition for convenience [19], and this is what we focus on here. The original definition in Lundberg and Lee [23] was actually the conditional one (although they apply the marginal definition in experiments, again for tractability), and recent work has studied approximations to compute it [1]. We expect that similar results to Theorem 3.3 could be proved in this setting, but this would require a different proof technique and hence we defer it to future work.

Definition A.1 (SHAP [23]). *Let*

$$\omega(i) = \frac{i!(p-i-1)!}{p!}.$$

For all $f \in \mathcal{F}$, $\mu \in \mathcal{P}(\mathcal{X})$, $x \in \mathcal{X}$, and $j \in [p]$, define

$$\Phi^{\text{SHAP}}(f, \mu, x)_j = \mathbb{E}_{X \sim \mu} \left[\sum_{\mathcal{S} \subseteq [p]} \omega(|\mathcal{S}|) \left(f(x_{\mathcal{S} \cup \{j\}}, X_{\mathcal{S}^c \setminus \{j\}}) - f(x_{\mathcal{S}}, X_{\mathcal{S}^c}) \right) \right].$$

For Integrated Gradients, it is usually only defined with a single baseline example. In order to state our results most generally, we consider the natural extension of averaging over a baseline distribution, which recovers the original definition when the baseline is a pointmass.

Definition A.2 (Integrated Gradients [41]). *For all $f \in \mathcal{F}$, $\mu \in \mathcal{P}(\mathcal{X})$, $x \in \mathcal{X}$, and $j \in [p]$, whenever f is appropriately differentiable define*

$$\Phi^{\text{IG}}(f, \mu, x)_j = \mathbb{E}_{X \sim \mu} \left[(x_j - X_j) \int_0^1 \nabla_j f(X + \alpha(x - X)) d\alpha \right].$$

Next, we have the precise statements and proofs of the informal claims at the start of Section 3: SHAP and Integrated Gradients are both complete and linear.

Proposition A.3. *SHAP is complete.*

Proof of Proposition A.3. For all $x, x' \in \mathcal{X}$,

$$\begin{aligned} & \sum_{j \in [p]} \sum_{\mathcal{S} \subseteq [p]} \omega(|\mathcal{S}|) \left(f(x_{\mathcal{S} \cup \{j\}}, x'_{\mathcal{S}^c \setminus \{j\}}) - f(x_{\mathcal{S}}, x'_{\mathcal{S}^c}) \right) \\ &= \sum_{\mathcal{S} \subseteq [p]} \sum_{j \in \mathcal{S}} \omega(|\mathcal{S}| - 1) f(x_{\mathcal{S}}, x'_{\mathcal{S}^c}) - \sum_{\mathcal{S} \subseteq [p]} \sum_{j \notin \mathcal{S}} \omega(|\mathcal{S}|) f(x_{\mathcal{S}}, x'_{\mathcal{S}^c}) \\ &= \sum_{\mathcal{S} \subseteq [p]} f(x_{\mathcal{S}}, x'_{\mathcal{S}^c}) \left(|\mathcal{S}| \omega(|\mathcal{S}| - 1) - (p - |\mathcal{S}|) \omega(|\mathcal{S}|) \right) \\ &= p \cdot \omega(p - 1) f(x) - p \cdot \omega(0) f(x') + \sum_{\mathcal{S} \subseteq [p], \mathcal{S} \neq \emptyset, [p]} f(x_{\mathcal{S}}, x'_{\mathcal{S}^c}) \left(|\mathcal{S}| \omega(|\mathcal{S}| - 1) - (p - |\mathcal{S}|) \omega(|\mathcal{S}|) \right). \end{aligned}$$

Then, observe that

$$\omega(p-1) = \omega(0) = \frac{1}{p},$$

and for $0 < i < p$,

$$\begin{aligned} i \cdot \omega(i-1) &= i \frac{(i-1)!(p-i)!}{p!} = \frac{i!(p-i)!}{p!} \quad \text{and} \\ (p-i) \cdot \omega(i) &= (p-i) \frac{i!(p-i-1)!}{p!} = \frac{i!(p-i)!}{p!}. \end{aligned}$$

The result follows by replacing x' with $X \sim \mu$ and taking expectation. $\square$

Proposition A.4 ([21]). *If for all $f \in \mathcal{F}$, either (a) the f is continuously differentiable everywhere; or (b) $\mathcal{X}$ is open, there exists L such that f is L -Lipschitz, and for all $x, \bar{x} \in \mathcal{X}$ and almost every $\alpha \in [0, 1]$, f is differentiable at $\bar{x} + \alpha(x - \bar{x})$, then Integrated Gradients is complete.*

Remark A.5. *This is a weaker statement than Proposition 1 of Sundararajan et al. [41], which Lerma and Lucas [21] show is false in general.* $\triangleleft$

Proposition A.6. *Integrated Gradients and SHAP are linear.*

Proof of Proposition A.6. Fix $f_1, \dots, f_p : \mathbb{R} \rightarrow \mathcal{Y}$ and $f(x) = \sum_{j \in [p]} f_j(x_j)$, $\mu \in \mathcal{P}(\mathcal{X})$, and $x \in \mathcal{X}$.

For Integrated Gradients, for all $\bar{x} \in \mathcal{X}$ and $j \in [p]$,

$$\begin{aligned} \Phi^{\text{IG}}(f, \bar{x}, x)_j &= (x_j - \bar{x}_j) \int_0^1 \nabla_j f(\bar{x} + \alpha(x - \bar{x})) \, d\alpha \\ &= (x_j - \bar{x}_j) \int_0^1 \nabla_j \sum_{\ell \in [p]} f_\ell(\bar{x}_\ell + \alpha(x_\ell - \bar{x}_\ell)) \, d\alpha \\ &= (x_j - \bar{x}_j) \int_0^1 \nabla_j f_j(\bar{x}_j + \alpha(x_j - \bar{x}_j)) \, d\alpha \\ &= \Phi^{\text{IG}}(f_j, \bar{x}_j, x_j). \end{aligned}$$

The result follows from taking expectation under μ .

For SHAP,

$$\begin{aligned} \Phi^{\text{SHAP}}(f, \mu, x)_j &= \mathbb{E}_{X \sim \mu} \left[\sum_{S \subseteq [p]} \omega(|S|) \left(f(x_{S \cup \{j\}}, X_{S^c \setminus \{j\}}) - f(x_S, X_{S^c}) \right) \right] \\ &= \mathbb{E}_{X \sim \mu} \left[\sum_{S \subseteq [p] : j \notin S} \omega(|S|) \left(\sum_{i \in S} f_i(x_i) + f_j(x_j) + \sum_{\ell \in S^c \setminus \{j\}} f_\ell(X_\ell) \right. \right. \\ &\quad \left. \left. - \sum_{i \in S} f_i(x_i) - f_j(X_j) - \sum_{\ell \in S^c \setminus \{j\}} f_\ell(X_\ell) \right) \right] \\ &= f_j(x_j) - \mathbb{E}_{X \sim \mu} f_j(X_j) \\ &= \Phi^{\text{SHAP}}(f_j, \mu_j, x_j), \end{aligned}$$

where the last step holds since SHAP is complete. $\square$

A.1 Additional Feature Attribution Methods

In Section 4, we compare SHAP and Integrated Gradients to other feature attribution methods that *do not* satisfy completeness and linearity. To keep the paper self-contained, we redefine these here.

Definition A.7 (SmoothGrad [37]). *For all $f \in \mathcal{F}$, $\mu \in \mathcal{P}(\mathcal{X})$, $x \in \mathcal{X}$, and $j \in [p]$, whenever f is appropriately differentiable define*

$$\Phi^{\text{SG}}(f, \mu, x)_j = \mathbb{E}_{x' \sim \mu} \nabla_j f(x').$$

Definition A.8 (LIME [33]). *Fix $\lambda > 0$. For all $f \in \mathcal{F}$, $\mu \in \mathcal{P}(\mathcal{X})$, and $x \in \mathcal{X}$, define*

$$\Phi^{\text{LIME}}(f, \mu, x) = \arg \min_{\beta \in \mathbb{R}^p} \left[\mathbb{E}_{x' \sim \mu} \left(\langle \beta, x' \rangle - f(x') \right)^2 + \lambda \|\beta\|_2^2 \right].$$

B Proofs for Impossibility Results

B.1 Generalized Assumptions

First, we restate Assumptions 1 and 2 in the multivariate case to allow for the most general version of our results.

Assumption B.1. *For any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, and $\mu \in \mathcal{P}(\mathcal{X})$, we say the present assumption holds if Assumption 1 holds for each $j \in \mathcal{S}$ with δ_j .*

Assumption B.2. *For any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, and $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, let $\mathcal{B} = \prod_{j \in \mathcal{S}} [x_j - \delta_j, x_j + \delta_j] \times x_{[p] \setminus \mathcal{S}}$. For any $(g_j : [x_j - \delta_j, x_j + \delta_j] \rightarrow \mathcal{Y})_{j \in \mathcal{S}}$ and $m \in \mathbb{N}$, we say that the present assumption holds with size m if*

$$\left\{ f \in (\mathcal{X} \rightarrow \mathcal{Y}) : \forall x' \in \mathcal{B} \quad f(x') = \sum_{j \in \mathcal{S}} g_j(x'_j), \quad f|_{\mathcal{B}^c} \in \mathcal{L}^m|_{\mathcal{B}^c}, \quad f \text{ is continuous} \right\} \subseteq \mathcal{F}.$$

B.2 Proof of Main Result

We state and prove the following multivariate generalization.

Theorem B.3. *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, $\mu \in \mathcal{P}(\mathcal{X})$, and $(g_j^{(0)}, g_j^{(1)} : [x_j - \delta_j, x_j + \delta_j] \rightarrow \mathcal{Y})_{j \in \mathcal{S}}$. Suppose that Assumption B.1 is satisfied and Assumption B.2 is satisfied with $m = 2^{|\mathcal{S}|}$. Define $\mathcal{B}$ as in Assumption B.2, and let*

$$\begin{aligned} \mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \forall x' \in \mathcal{B}, \quad f(x') = \sum_{j \in \mathcal{S}} g_j^{(0)}(x'_j) \right\} \\ \mathcal{F}^{(1)} &= \left\{ f \in \mathcal{F} : \forall x' \in \mathcal{B}, \quad f(x') = \sum_{j \in \mathcal{S}} g_j^{(1)}(x'_j) \right\}. \end{aligned}$$

For any complete and linear Φ and feature-attribution hypothesis test $\mathbf{h}$,

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

Before we prove this, we need the following technical result, which is the multivariate analogue of Theorem 3.4.

Theorem B.4. *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, $\mu \in \mathcal{P}(\mathcal{X})$, and $(g_j : [x_j - \delta_j, x_j + \delta_j] \rightarrow \mathcal{Y})_{j \in \mathcal{S}}$. Suppose that Assumption B.1 is satisfied and Assumption B.2 is satisfied with $m = 2^{|\mathcal{S}|}$. For every $\phi \in \mathbb{R}^{|\mathcal{S}| \times q}$, there exists $f \in \mathcal{F}$ such that for every complete and linear feature attribution method, $\Phi(f, \mu, x)_{\mathcal{S}} = \phi$, and if $|x_j - x'_j| \leq \delta_j$ for all $j \in \mathcal{S}$ then $f(x') = \sum_{j \in \mathcal{S}} g_j(x'_j)$.*

We are now able to prove our main theorem.

Proof of Theorem B.3. For $j \notin \mathcal{S}$, let $g_j^{(0)} = g_j^{(1)} \equiv 0$. Fix $\phi = \mathbf{0}$. Let $f^{(0)}$ be the model guaranteed to exist from Theorem B.4 for $(g_j^{(0)})_{j \in [p]}$, and similarly define $f^{(1)}$. Since $f^{(0)}, f^{(1)} \in \mathcal{F}$ by Assumption B.2,

$$\begin{aligned} \text{Spec}_{\Phi, \mu, x}(\mathbf{h}) &= \inf_{f \in \mathcal{F}^{(0)}} [1 - \mathbf{h}(\Phi(f, \mu, x))] \\ &\leq 1 - \mathbf{h}(\Phi(f^{(0)}, \mu, x)) \\ &= 1 - \mathbf{h}(\mathbf{0}) \\ &= 1 - \mathbf{h}(\Phi(f^{(1)}, \mu, x)) \\ &\leq 1 - \inf_{f \in \mathcal{F}^{(1)}} \mathbf{h}(\Phi(f, \mu, x)) \\ &= 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}). \end{aligned}$$

□

B.3 Proof of Theorem B.4

The proof follows by explicitly constructing a piecewise model f that satisfies the desired properties and is linear outside of the neighbourhood around x . In particular, for each $j \in \mathcal{S}$ and $k \in [q]$, we construct f_{jk}^L and f_{jk}^R , define

$$f_{jk}(x'_j) = \begin{cases} f_{jk}^L(x'_j) & x'_j \in (x_j^L, x_j - \delta_j) \\ g_j(x'_j)_k & x'_j \in [x_j - \delta_j, x_j + \delta_j] \\ f_{jk}^R(x'_j) & x'_j \in (x_j + \delta_j, x_j^R) \\ 0 & \text{otherwise,} \end{cases}$$

and finally define

$$f(x')_k = \sum_{j \in \mathcal{S}} f_{jk}(x'_j).$$

By definition, $f(x') = \sum_{j \in \mathcal{S}} g_j(x'_j)$ if $|x'_j - x_j| \leq \delta_j$ for all $j \in \mathcal{S}$. Further, since Φ is linear,

$$\Phi(f, x, \mu)_{jk} = \Phi(f_{jk}, x_j, \mu_j).$$

For some $\beta_{jk}^L, \beta_{jk}^R \in \mathbb{R}$ to be chosen in the proof, set

$$f_{jk}^L(x'_j) = \beta_{jk}^L \cdot (x'_j - x_j + \delta_j) + g_j(x_j - \delta_j)_k$$

and

$$f_{jk}^R(x'_j) = \beta_{jk}^R \cdot (x'_j - x_j - \delta_j) + g_j(x_j + \delta_j)_k.$$

These functions are piecewise linear on $(x_j^L, x_j - \delta_j)$ and $(x_j + \delta_j, x_j^R)$ for each $j \in \mathcal{S}$ respectively, and hence f is piecewise linear on the product of these intervals (which form convex polytopes). Thus, by Assumption 2, $f \in \mathcal{F}$. It remains to construct β_{jk}^L and $\beta_{jk}^R \in \mathbb{R}$ so that $\Phi(f_{jk}, x_j, \mu_j) = \phi_{jk}$ for each $j \in \mathcal{S}$ and $k \in [q]$.

Fix $j \in [p]$ and $k \in [q]$. Recall that μ_j is a probability measure, and hence maps intervals of the form (a, b) to a number in $[0, 1]$.

For any $\beta_{jk}^L, \beta_{jk}^R \in \mathbb{R}$ that satisfy

$$\begin{aligned} & \beta_{jk}^L \left[\mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j^L, x_j - \delta_j)\}] - (x_j - \delta_j) \cdot \mu_j((x_j^L, x_j - \delta_j)) \right] \\ & + \beta_{jk}^R \left[\mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j + \delta_j, x_j^R)\}] - (x_j + \delta_j) \cdot \mu_j((x_j + \delta_j, x_j^R)) \right] \\ & = -g_j(x_j - \delta_j)_k \cdot \mu_j((x_j^L, x_j - \delta_j)) - g_j(x_j + \delta_j)_k \cdot \mu_j((x_j + \delta_j, x_j^R)) \\ & \quad - \mathbb{E}_{X_j \sim \mu_j} [g_j(X_j)_k \cdot \mathbb{I}\{X_j \in [x_j - \delta_j, x_j + \delta_j]\}] + g_j(x_j)_k - \phi_{jk}, \end{aligned}$$

since Φ is complete,

$$\Phi(f_{jk}, x_j, \mu_j) = g_j(x_j)_k - \mathbb{E}_{X_j \sim \mu_j} [f_{jk}(X_j)] = \phi_{jk}.$$

For simplicity, we set either β_{jk}^L or β_{jk}^R to zero (at worst, this inflates the Lipschitz parameter by a factor of 2). By assumption (i),

$$\max \left\{ \mu_j((x_j^L, x_j - \delta_j)), \mu_j((x_j + \delta_j, x_j^R)) \right\} > 0,$$

so one of

$$\mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j^L, x_j - \delta_j)\}] - (x_j - \delta_j) \cdot \mu_j((x_j^L, x_j - \delta_j))$$

and

$$\mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j + \delta_j, x_j^R)\}] - (x_j + \delta_j) \cdot \mu_j((x_j + \delta_j, x_j^R))$$

are non-zero. If

$$\begin{aligned} & \left| \mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j^L, x_j - \delta_j)\}] - (x_j - \delta_j) \cdot \mu_j((x_j^L, x_j - \delta_j)) \right| \\ & > \left| \mathbb{E}_{X_j \sim \mu_j} [X_j \cdot \mathbb{I}\{X_j \in (x_j + \delta_j, x_j^R)\}] - (x_j + \delta_j) \cdot \mu_j((x_j + \delta_j, x_j^R)) \right|, \end{aligned}$$

set $\beta_{jk}^R = 0$, and otherwise set $\beta_{jk}^L = 0$. □

C Proofs for End-Tasks

C.1 Proof of Proposition 3.5

Since $\nabla f(x)$ exists,

$$\nabla_j f(x) = \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha e_j) - f(x)}{\alpha}.$$

Thus, for any $\varepsilon' > 0$ there exists $\delta > 0$ such that for all $\alpha \leq \delta$,

$$\begin{aligned} \frac{f(x + \alpha e_j) - f(x)}{\alpha} &\in [\alpha \nabla_j f(x) - \alpha \varepsilon/4, \alpha \nabla_j f(x) + \alpha \varepsilon/4] \\ &\subseteq [\alpha \nabla_j f(x) - \delta \varepsilon/4, \alpha \nabla_j f(x) + \delta \varepsilon/4] \\ &:= S(\alpha). \end{aligned}$$

Let $\mathbf{h} = 1 - \mathbb{I}\{\sup_{\alpha \leq \delta} \sup_{b \in S(\alpha)} |b| \leq \delta \varepsilon\}$. When $f \in \mathcal{F}^{(1)}$, then there exists $\alpha \leq \delta$ such that for some $b \in S(\alpha)$, $|b| = |f(x + \alpha e_j) - f(x)| > \delta \varepsilon$, and thus $\mathbf{h} = 1$. Similarly, when $f \in \mathcal{F}^{(0)}$, then for every $\alpha \leq \delta$ and every $b \in S(\alpha)$,

$$|b| \leq |f(x + \alpha e_j) - f(x)| + \delta \varepsilon/2 \leq \delta \varepsilon.$$

This implies that $\mathbf{h} = 0$.

□

C.2 Proof of Proposition 3.6

Define $g^{(0)} \equiv 0$ and $g^{(1)}(x'_j) = x'_j - x_j$, and let $\mathcal{G}^{(0)}$ and $\mathcal{G}^{(1)}$ denote $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$ from Theorem 3.3 for the δ specified in the statement of Proposition 3.6. Clearly, $\mathcal{G}^{(0)} \subseteq \mathcal{F}^{(0)}$ and $\mathcal{G}^{(1)} \subseteq \mathcal{F}^{(1)}$ as respectively defined in Proposition 3.6. For $\ell \neq j$, let $g_\ell^{(0)} = g_\ell^{(1)} \equiv 0$. By the assumption on $\mathcal{F}$ in Proposition 3.6, Assumption 2 is satisfied. Thus, by Theorem 3.3,

$$\begin{aligned} \text{Spec}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^{(0)}) &= \inf_{f \in \mathcal{F}^{(0)}} [1 - \mathbf{h}(\Phi(f, \mu, x))] \\ &\leq \inf_{f \in \mathcal{G}^{(0)}} [1 - \mathbf{h}(\Phi(f, \mu, x))] \\ &= \text{Spec}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{G}^{(0)}) \\ &\leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{G}^{(1)}) \\ &= 1 - \inf_{f \in \mathcal{G}^{(1)}} \mathbf{h}(\Phi(f, \mu, x)) \\ &\leq 1 - \inf_{f \in \mathcal{F}^{(1)}} \mathbf{h}(\Phi(f, \mu, x)) \\ &= 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^{(1)}). \end{aligned}$$

□

C.3 Proof of Proposition 3.10

By completeness and linearity, taking $x = 1$ gives $\Phi(f, \mu, x) = a(x^n - \mathbb{E}_\mu X^n) - (x - \mathbb{E}_\mu X) = a(1 - \mathbb{E}_\mu X^n) - 1$ while $f'(x) = nax^{n-1} - 1$. Thus, $\Phi(f, \mu, x) > 0$ if and only if $a > 1/(1 - \mathbb{E}_\mu X^n)$

while $f'(x) > 0$ if and only if $a > 1/n$. This implies that

$$\begin{aligned}\mathbb{P}_{f \sim \pi} \left[\text{sgn}(\Phi(f, \mu, x)) \neq \text{sgn}(f'(x)) \right] &= \mathbb{P}_{a \sim \text{Gaussian}(0,1)} [a \in (1/n, 1/(1 - \mathbb{E}_\mu X^n))] \\ &\geq \mathbb{P}_{a \sim \text{Gaussian}(0,1)} [a \in (1/2, 2)] \\ &\approx 0.2858,\end{aligned}$$

where the approximation is of the Gaussian CDF. The upper bound is because $\mathbb{P}_{a \sim \text{Gaussian}(0,1)} [a > 1/n] \leq 0.5$. $\square$

C.4 Proofs for Recourse and Spurious Features

We first state the multivariate versions of these definitions for the most generality.

Definition C.1 (Recourse). *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $k \in [q]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, and $\nu \in \mathcal{P}(\mathcal{X})$. For any $\sigma \in \{\pm 1\}^{|\mathcal{S}|}$, let $\text{Recourse}_{\mathcal{S}k}(x, \nu, \delta, \sigma)$ be defined by*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \mathbb{E}_{X \sim \nu} \left[f(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j + \sigma_j \delta_j] \right] \right. \\ &\quad \left. > \mathbb{E}_{X \sim \nu} \left[f(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j - \sigma_j \delta_j] \right] \right\} \\ \mathcal{F}^{(1)} &= \mathcal{F} \setminus \mathcal{F}^{(0)}.\end{aligned}$$

Definition C.2 (Spurious Features). *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $k \in [q]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, and $\varepsilon > 0$. For any $\sigma \in \{\pm 1\}^{|\mathcal{S}|}$, let $\text{Spurious}_{\mathcal{S}k}(x, \delta, \sigma, \varepsilon)$ be defined by*

$$\begin{aligned}\mathcal{F}^{(0)} &= \left\{ f \in \mathcal{F} : \sup_{x' \in \prod_{j \in \mathcal{S}} (x_j, x_j + \sigma_j \delta_j] \times x_{[p] \setminus \mathcal{S}}} f(x')_k \leq 0 \right\} \\ \mathcal{F}^{(1)} &= \left\{ f \in \mathcal{F} : \sup_{x' \in \prod_{j \in \mathcal{S}} (x_j, x_j + \sigma_j \delta_j] \times x_{[p] \setminus \mathcal{S}}} f(x')_k \geq \varepsilon \right\}.\end{aligned}$$

Corollary C.3 (Generalization of Theorem 3.3 for Recourse). *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, and $\mu \in \mathcal{P}(\mathcal{X})$ such that Assumption B.1 is satisfied for each $j \in \mathcal{S}$. Fix $k \in [q]$, $\nu \in \mathcal{P}(\mathcal{X})$, and $\sigma \in \{\pm 1\}^{|\mathcal{S}|}$, and let $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$ be defined by $\text{Recourse}_{\mathcal{S}k}(x, \nu, \delta, \sigma)$. Suppose that there exists $f^{(0)} \in \mathcal{F}^{(0)}$ and $f^{(1)} \in \mathcal{F}^{(1)}$ that each are of the form $\sum_{j \in \mathcal{S}} g_j$ and each satisfy Assumption B.2. Then for any complete and linear feature attribution method Φ and feature-attribution hypothesis test $\mathbf{h}$,*

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

Proof of Corollary C.3. This proof will benefit from slightly more detailed notation for Spec and Sens. Specifically, we denote the explicit dependence on $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$. Let $\mathcal{F}^L$ and $\mathcal{F}^R$ denote $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$ respectively from Definition C.1. Pick the $g^{(0)} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{aligned}\mathbb{E}_{X \sim \nu} \left[g^{(0)}(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j + \sigma_j \delta_j] \right] \\ > \mathbb{E}_{X \sim \nu} \left[g^{(0)}(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j - \sigma_j \delta_j] \right]\end{aligned}$$

as prescribed in the statement of Corollary C.3. Let $\mathcal{F}^{(0)}$ be defined as in Theorem 3.3 for $g^{(0)}$. Clearly, $\mathcal{F}^{(0)} \subseteq \mathcal{F}^L$. Similarly, pick the $g^{(1)} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \mathbb{E}_{X \sim \nu} [g^{(1)}(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j + \sigma_j \delta_j]] \\ \leq \mathbb{E}_{X \sim \nu} [g^{(1)}(x_{[p] \setminus \mathcal{S}}, X_{\mathcal{S}})_k \mid X_{\mathcal{S}} \in \prod_{j \in \mathcal{S}} [x_j, x_j - \sigma_j \delta_j]] \end{aligned}$$

as prescribed in the statement. Define $\mathcal{F}^{(1)}$ accordingly, and observe that $\mathcal{F}^{(1)} \subseteq \mathcal{F}^R$. By assumption, Assumption B.2 is satisfied, and thus we can apply Theorem B.3 to obtain

$$\begin{aligned} \text{Spec}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^L, \mathcal{F}^R) &= \inf_{f \in \mathcal{F}^L} [1 - \mathbf{h}(\Phi(f, \mu, x))] \\ &\leq \inf_{f \in \mathcal{F}^{(0)}} [1 - \mathbf{h}(\Phi(f, \mu, x))] \\ &= \text{Spec}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^{(0)}, \mathcal{F}^{(1)}) \\ &\leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^{(0)}, \mathcal{F}^{(1)}) \\ &= 1 - \inf_{f \in \mathcal{F}^{(1)}} \mathbf{h}(\Phi(f, \mu, x)) \\ &\leq 1 - \inf_{f \in \mathcal{F}^R} \mathbf{h}(\Phi(f, \mu, x)) \\ &= 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}; \mathcal{F}^L, \mathcal{F}^R). \end{aligned}$$

□

Corollary C.4 (Generalization of Theorem 3.3 for Spurious Features). *Fix any $x \in \mathcal{X}$, $\mathcal{S} \subseteq [p]$, $\delta \in \mathbb{R}_+^{|\mathcal{S}|}$, and $\mu \in \mathcal{P}(\mathcal{X})$ such that Assumption B.1 is satisfied for each $j \in \mathcal{S}$. Fix $k \in [q]$, $\varepsilon > 0$, and $\sigma \in \{\pm 1\}^{|\mathcal{S}|}$, and let $\mathcal{F}^{(0)}$ and $\mathcal{F}^{(1)}$ be defined by $\text{Spurious}_{\mathcal{S}_k}(x, \delta, \sigma, \varepsilon)$. Suppose that there exists $f^{(0)} \in \mathcal{F}^{(0)}$ and $f^{(1)} \in \mathcal{F}^{(1)}$ that each are of the form $\sum_{j \in \mathcal{S}} g_j$ and each satisfy Assumption B.2. Then, for any complete and linear feature attribution method Φ and feature-attribution hypothesis test $\mathbf{h}$,*

$$\text{Spec}_{\Phi, \mu, x}(\mathbf{h}) \leq 1 - \text{Sens}_{\Phi, \mu, x}(\mathbf{h}).$$

Proof of Corollary C.4. The proof is identical to that of Corollary C.3, just with $g^{(0)}$ and $g^{(1)}$ defined appropriately to match the behaviour described by Definition C.2 (e.g., constant 0 and constant ε would work). □

D Proofs for Brute-Force Model Evaluations

D.1 Notation for Sample Complexity

We begin with additional notation to formalize brute-force model evaluations. A *query algorithm* is any way to sequentially choose examples to query the model at. Let $\mathcal{Q} = \mathcal{X} \times \mathcal{Y} \times \mathbb{R}^{p \times q}$. Formally, a query algorithm is any sequence of functions $\mathbf{q} = (\mathbf{q}^{(t)})_{t \in \mathbb{N}}$ such that for each t ,

$$\mathbf{q}^{(t)} : \mathcal{Q}^{t-1} \rightarrow \mathcal{P}(\mathcal{X}).$$

For any $f \in \mathcal{F}$, we write $X^{(1:n)} \sim \mathbf{q}^{(1:n)}[f]$ if

$$\begin{aligned} X^{(1)} &\sim \mathbf{q}^{(1)} \quad \text{and} \\ \forall t > 1 \quad X^{(t)} &\mid \left(X^{(1:t-1)}, f(X^{(1:t-1)}), \nabla f(X^{(1:t-1)}) \right) \sim \mathbf{q} \left(X^{(1:t-1)}, f(X^{(1:t-1)}), \nabla f(X^{(1:t-1)}) \right). \end{aligned}$$

We now introduce *query hypothesis testing*, where the user makes a decision using only (random) queries of a feature attribution method. In practice, this is more general than feature-attribution hypothesis testing, since a user can only access the output of a feature attribution method via queries. Formally, a query hypothesis test is any function

$$\mathbf{s} : 2^{\mathcal{Q}} \rightarrow [0, 1].$$

This time, the output of $\mathbf{s}$ is *the probability that the user rejects the null hypothesis, conditional on the observed queries*. To study specificity and sensitivity for the entire procedure of selecting queries and then performing a query hypothesis test, we introduce the $\{0, 1\}$ -valued random variable $\mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)}$, which has the distribution defined by

$$\begin{aligned} \mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)} &\mid X^{(1:n)} \sim \text{Bernoulli} \left(\mathbf{s} \left(X^{(1:n)}, f(X^{(1:n)}), \nabla f(X^{(1:n)}) \right) \right) \\ X^{(1:n)} &\sim \mathbf{q}^{(1:n)}[f]. \end{aligned}$$

Finally, for a fixed $\mathbf{q}$, $\mathbf{s}$, and $\mathcal{F}^{(0)}, \mathcal{F}^{(1)} \subseteq \mathcal{F}$,

$$\begin{aligned} \text{Spec}^{(n)}(\mathbf{q}, \mathbf{s}) &= \inf_{f \in \mathcal{F}^{(0)}} \left[1 - \mathbb{E} \left[\mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)} \right] \right] \quad \text{and} \\ \text{Sens}^{(n)}(\mathbf{q}, \mathbf{s}) &= \inf_{f \in \mathcal{F}^{(1)}} \mathbb{E} \left[\mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)} \right]. \end{aligned}$$

Once again, the user's goal is to maximize these simultaneously.

D.2 Sample Complexity Theorems

We define the *Lipschitz constant* for any $f : \mathcal{X} \rightarrow \mathcal{Y}$, $\mathcal{B} \subseteq \mathcal{X}$, and $k \in [q]$ by

$$\text{Lip}^{\mathcal{B}}(f)_k = \sup_{x, x' \in \mathcal{B}} \frac{|f(x)_k - f(x')_k|}{\|x - x'\|_{\infty}}.$$

First, we show that the task shown to be impossible for feature-attribution hypothesis tests in Corollary C.4 can be solved with sufficiently many queries. For any $\delta > 0$, let $\mathbf{q}_{\delta}$ be such that

$$\mathbf{q}_{\delta}^{(t)} \equiv \text{Unif}((0, \delta]^p).$$

Further, for any $\tau \in [0, 1]$, define

$$\mathbf{s}_{\tau}^*(x^{(1:n)}, y^{(1:n)}) = \tau + (1 - \tau) \cdot \mathbb{I}\{\exists t \in [n] \text{ s.t. } y^{(t)} > 0\}.$$

Then, we have the following formal version of Theorem 5.1.

Theorem D.1. Fix arbitrary $\delta, \varepsilon > 0$ and $L > 0$. Suppose that for all $f \in \mathcal{F}$, $\text{Lip}^{(0,\delta]^p}(f) \leq L$, and let

$$\begin{aligned}\mathcal{F}^{(0)} &= \{f \in \mathcal{F} : \sup_{x \in (0,\delta]^p} f(x) \leq 0\} \\ \mathcal{F}^{(1)} &= \left\{f \in \mathcal{F} : \sup_{x \in (0,\delta]^p} f(x) \geq \varepsilon\right\}.\end{aligned}$$

Then, for any $\tau \in [0, 1]$,

$$\begin{aligned}\text{Spec}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) &= 1 - \tau \quad \text{and} \\ \text{Sens}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) &= 1 - (1 - \tau) \left(1 - \left(\frac{2\varepsilon}{L\delta}\right)^p\right)^n.\end{aligned}$$

In particular, this implies that

$$\text{Spec}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) = 1 - \text{Sens}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) + (1 - \tau) \cdot \left(1 - \left(1 - \left(\frac{2\varepsilon}{L\delta}\right)^p\right)^n\right).$$

The following result shows that this rate is nearly tight (as can be seen by a first-order Taylor expansion of $(1 - x)^n$).

Theorem D.2. In the same setting as Theorem D.1, if Assumption B.2 holds for linear g then for any model agnostic $\mathbf{q}, \mathbf{s}$, and $n \in \mathbb{N}$,

$$\text{Spec}^{(n)}(\mathbf{q}, \mathbf{s}) \leq 1 - \text{Sens}^{(n)}(\mathbf{q}, \mathbf{s}) + n \cdot \left(\left\lfloor \frac{L\delta}{2\varepsilon} \right\rfloor\right)^{-p}.$$

D.3 Proof of Theorem D.1

First, for any $f^{(0)} \in \mathcal{F}^{(0)}$ and $x^{(1:n)}$, observe that

$$\mathbb{E}[\mathbf{T}_{\mathbf{q}_\delta[f^{(0)}], \mathbf{s}_\tau^*}^{(n)} \mid x^{(1:n)}] = \mathbf{s}_\tau^*(x^{(1:n)}, f^{(0)}(x^{(1:n)})) = \tau + (1 - \tau) \cdot \mathbb{I}\{\exists t \in [n] \text{ s.t. } f^{(0)}(x^{(t)}) > 0\} = \tau.$$

That is,

$$\text{Spec}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) = \inf_{f \in \mathcal{F}^{(0)}} \left[1 - \mathbb{E}[\mathbf{T}_{\mathbf{q}_\delta[f], \mathbf{s}_\tau^*}^{(n)}]\right] = 1 - \tau.$$

Next, fix an arbitrary $f^{(1)} \in \mathcal{F}^{(1)}$. By definition, there exists $x^* \in (0, \delta]^p$ such that $f^{(1)}(x^*) \geq \varepsilon$. Since $\text{Lip}^{(0,\delta]^p}(f^{(1)}) \leq L$, if $\|x - x^*\|_\infty < \varepsilon/L$ then $f^{(1)}(x) > 0$. Let $h^* = \{x \in (0, \delta]^p : \|x - x^*\|_\infty < \varepsilon/L\}$,

$$\mathcal{A}^{(t)} = \{X^{(t)} \in h^*\}, \quad \text{and} \quad \mathcal{A} = \bigcup_{t \in [n]} \mathcal{A}^{(t)}.$$

Further, let $r = L\delta/(2\varepsilon)$. By definition, since $\mathbf{q}_\delta$ is independent of $f^{(1)}$,

$$\mathbb{P}_{\mathbf{q}_\delta^{(1:n)}}[\mathcal{A}^c] = (1 - r^{-p})^n.$$

Then,

$$\begin{aligned}
\text{Sens}^{(n)}(\mathbf{q}_\delta, \mathbf{s}_\tau^*) &= \inf_{f \in \mathcal{F}^{(1)}} \mathbb{E}[\mathbf{T}_{\mathbf{q}_\delta[f], \mathbf{s}_\tau^*}^{(n)}] \\
&= \inf_{f \in \mathcal{F}^{(1)}} \left(\mathbb{E}[\mathbf{T}_{\mathbf{q}_\delta[f], \mathbf{s}_\tau^*}^{(n)} \mid \mathcal{A}] \mathbb{P}_{\mathbf{q}_\delta^{(1:n)}}[\mathcal{A}] + \mathbb{E}[\mathbf{T}_{\mathbf{q}_\delta[f], \mathbf{s}_\tau^*}^{(n)} \mid \mathcal{A}^c] \mathbb{P}_{\mathbf{q}_\delta^{(1:n)}}[\mathcal{A}^c] \right) \\
&= \mathbb{P}_{\mathbf{q}_\delta^{(1:n)}}[\mathcal{A}] + \tau \cdot \mathbb{P}_{\mathbf{q}_\delta^{(1:n)}}[\mathcal{A}^c] \\
&= 1 - (1 - \tau)(1 - r^{-p})^n.
\end{aligned}$$

□

D.4 Proof of Theorem D.2

Let $f^{(0)} \equiv 0$ and $\mathbb{P}^{(0)}$ denote $\mathbb{P}_{X^{(1:n)} \sim \mathbf{q}^{(1:n)}[f^{(0)}]}$. Define $\lfloor L\delta/(2\varepsilon) \rfloor = r \in \mathbb{N}$. Divide $(0, r \cdot (2\varepsilon/L)]^p$ into r^p cubes (each denoted by h_ℓ for $\ell \in [r^p]$) with sides that are left-open and right-closed of length $2\varepsilon/L$. For each $\ell \in [r^p]$, let

$$\mathcal{A}_\ell^{(t)} = \{X^{(t)} \in h_\ell\}$$

and

$$\mathcal{A}_\ell = \bigcup_{t \in [n]} \mathcal{A}_\ell^{(t)}.$$

By a union bound,

$$\mathbb{P}^{(0)}[\mathcal{A}_\ell] \leq \sum_{t \in [n]} \mathbb{P}^{(0)}[\mathcal{A}_\ell^{(t)}].$$

Further, since $h_\ell \cap h_{\ell'} = \emptyset$ for all $\ell \neq \ell'$ and

$$\bigcup_{\ell \in [r^p]} h_\ell \subseteq (0, \delta]^p,$$

$$\sum_{\ell \in [r^p]} \sum_{t \in [n]} \mathbb{P}^{(0)}[\mathcal{A}_\ell^{(t)}] \leq n.$$

Thus, there must be some ℓ^* such that

$$\sum_{t \in [n]} \mathbb{P}^{(0)}[\mathcal{A}_{\ell^*}^{(t)}] \leq \frac{n}{r^p}.$$

Denote $\mathcal{A} = \mathcal{A}_{\ell^*}$ and $\mathcal{A}^{(t)} = \mathcal{A}_{\ell^*}^{(t)}$.

Define the $2L$ -Lipschitz bump function taking maximal value ε by

$$f^{(1)}(x) = \begin{cases} 0 & x \notin h_{\ell^*} \\ \text{square-based pyramid} & x \in h_{\ell^*}. \end{cases}$$

Observe that $f^{(1)} \in \mathcal{F}$ by Assumption 2. Similarly to $\mathbb{P}^{(0)}$, let $\mathbb{P}^{(1)}$ denote $\mathbb{P}_{X^{(1:n)} \sim \mathbf{q}^{(1:n)}[f^{(1)}]}$. Since $f^{(0)} = f^{(1)}$ outside of h_{ℓ^*} , it holds that

$$\mathbb{P}^{(0)}[\mathcal{A}] = \mathbb{P}^{(1)}[\mathcal{A}]$$

and

$$\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)} \stackrel{d}{=} \mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)} \mid \mathcal{A}^c.$$

Next,

$$\begin{aligned} \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)}] &= \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)} \mid \mathcal{A}^c] \mathbb{P}^{(1)}[\mathcal{A}^c] + \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)} \mid \mathcal{A}] \mathbb{P}^{(1)}[\mathcal{A}] \\ &= \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)} \mid \mathcal{A}^c] \mathbb{P}^{(0)}[\mathcal{A}^c] + \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)} \mid \mathcal{A}] \mathbb{P}^{(0)}[\mathcal{A}]. \end{aligned}$$

Similarly,

$$\mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)}] = \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)} \mid \mathcal{A}^c] \mathbb{P}^{(0)}[\mathcal{A}^c] + \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)} \mid \mathcal{A}] \mathbb{P}^{(0)}[\mathcal{A}].$$

Combining these, we obtain

$$\begin{aligned} \text{Spec}^{(n)}(\mathbf{q}, \mathbf{s}) &= \inf_{f \in \mathcal{F}^{(0)}} [1 - \mathbb{E}[\mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)}]] \\ &\leq 1 - \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)}] \\ &= 1 - \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)}] + \mathbb{P}^{(0)}[\mathcal{A}] (\mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(1)}], \mathbf{s}}^{(n)} \mid \mathcal{A}] - \mathbb{E}[\mathbf{T}_{\mathbf{q}[f^{(0)}], \mathbf{s}}^{(n)} \mid \mathcal{A}]) \\ &\leq 1 - \inf_{f \in \mathcal{F}^{(1)}} \mathbb{E}[\mathbf{T}_{\mathbf{q}[f], \mathbf{s}}^{(n)}] + \mathbb{P}^{(0)}[\mathcal{A}] \\ &= 1 - \text{Sens}^{(n)}(\mathbf{q}, \mathbf{s}) + \frac{n}{r^p}. \end{aligned}$$

□